Earnings-Call Text and the Dynamics of Uncertainty and Trading

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Abstract

Information discussed in earnings calls has been shown to predict price volatility in the days immediately following the call. In this paper, I extend the analysis to other well-established outcomes (cumulative abnormal returns, intraday realised-variance, abnormal volume) and various horizons following a call. Using modern machine learning techniques and recent advances in natural language processing, I assess the incremental forecasting improvement gained from including textual information from earnings calls alongside standard econometric controls. I show that the absorption speed of textual information differs by outcome, with its addition accounting for large improvements in forecasting performance for post-call realised variance at short horizons which quickly dissipates, weak but statistically significant improvements in abnormal return forecasting at medium horizons, and strong and prolonged improvement in abnormal volume at nearly all horizons. I further show that earnings-call text generates coordinated uncertainty and attention responses across outcomes: the text-driven increment to realised variance and abnormal volume are strongly co-moving (r = 0.69), while the text increment to abnormal returns is orthogonal to both. Moreover, text-driven volatility uncertainty at short horizons predicts subsequent abnormal trading activity for up to three weeks, with a clear decay structure consistent with gradual uncertainty resolution rather than noise rebalancing. The evidence is consistent with qualitative disclosure generating heterogeneous investor interpretation that produces a coordinated short-lived uncertainty shock, subsequently resolved through weeks of abnormal trading activity.

1. Introduction

Earnings announcements are among the most important scheduled information events in equity markets. They combine hard quantitative news, in the form of realised earnings relative to analyst expectations, with softer qualitative disclosure through the earnings conference call, where managers explain results, discuss guidance, and respond to analyst questions. How quickly these different forms of information are absorbed into prices, uncertainty, and trading activity is a central question in market efficiency and information processing. The post-earnings-announcement drift literature shows that even the hard earnings number is not always incorporated immediately into prices (Ball and Brown, 1968; Bernard and Thomas, 1989, 1990; Foster, Olsen and Shevlin, 1984; Livnat and Mendenhall, 2006). For qualitative disclosure, which is less standardised, harder to process, and more open to interpretation, the absorption process is less well understood.

This distinction matters because quantitative earnings news and qualitative disclosure are likely to affect markets through different channels. A hard earnings surprise provides investors with a relatively common signal about firm performance, while the same managerial explanation, risk discussion, or forward-looking statement may be interpreted differently by different investors, creating scope for disagreement even after the public disclosure has occurred. This implies that transcript information may matter less for directional return predictability, where heterogeneous views can offset one another in prices, and more for uncertainty, attention, and trading activity, where heterogeneous interpretation is itself part of the mechanism. Prior research shows that earnings calls contain information beyond the hard earnings release (Tasker, 1998; Frankel, Johnson and Skinner, 1999; Kimbrough, 2005), while textual-analysis studies link call tone, linguistic features, and disclosure language to market reactions (Loughran and McDonald, 2011; Price et al., 2012; Bochkay, Hales and Chava, 2020). What is less clear is how persistent this text-driven information is, and whether its persistence differs across prices, volatility, and trading activity.

This paper asks how long earnings-call transcripts retain incremental predictive value after the call, and whether that value differs across market outcomes. I study three outcomes: cumulative abnormal returns (CAR), intraday realised variance (RV), and abnormal trading volume (VOL). These outcomes capture distinct dimensions of market adjustment. Returns measure directional price discovery. Realised variance captures uncertainty and risk reassessment, which may rise when disclosure creates ambiguity even if the direction of the news is unclear. Abnormal volume captures attention, disagreement, and portfolio rebalancing, which theory predicts should respond strongly when investors interpret the same public signal differently (Harris and Raviv, 1993; Kandel and Pearson, 1995). By studying these outcomes jointly over horizons from one to sixty trading days (twenty-one trading days for realised variance), the paper characterises how earnings-call information is absorbed across distinct channels.

The empirical design is based on strict out-of-sample forecasting. Earnings-call transcripts are represented using a sentence-embedding model, **`BAAI/bge-base-en-v1.5`**, reduced using principal components estimated only on the training sample, and combined with hard earnings variables and outcome-specific baseline controls. The realised-variance models include HAR-style volatility controls; the abnormal-volume models include pre-event trading-volume controls; and the abnormal-return models include market-beta controls. For each outcome and horizon, I estimate nested specifications that separately identify the predictive contribution of baseline controls, hard earnings information, transcript text, and their combination. Models are trained and evaluated using a chronological 60/15/25 train-validation-test split. Forecast performance is assessed using out-of-sample R^2, Diebold-Mariano tests, and stationary-bootstrap confidence intervals, with Holm-Bonferroni adjustments for multiple testing. The design therefore asks whether transcript text improves predictions using only information available at the time of the call, evaluated on data not used in model selection or estimation.

The analysis uses more than 24,000 earnings-call transcripts for S&P 500 constituents from 2010 Q2 to 2025 Q2, matched to intraday market data, earnings-announcement information, and analyst expectations.

The results support a clear economic pattern. Hard earnings information provides essentially no incremental predictive power for abnormal returns once standard controls are included, consistent with relatively rapid price adjustment to the quantitative earnings news. Earnings-call text, however, materially improves forecasts of uncertainty and trading activity. For realised variance, transcript text generates large short-horizon improvements, with RMSE reductions of up to roughly 8–10% at one- to three-day horizons, before the incremental value decays by around two weeks. For abnormal volume, the effect is larger and more persistent, with RMSE reductions of approximately 15–18% at the one-day horizon and positive gains that remain visible even at longer horizons. For abnormal returns, by contrast, the text signal is statistically detectable but economically small, with RMSE improvements of around 0.1–0.3% concentrated at five- to ten-day horizons. The central empirical result is therefore that qualitative disclosure matters most for uncertainty and trading activity, and much less for directional return prediction.

A further contribution concerns the joint structure of the text signal across outcomes. At the event level, the text-driven increments to realised variance and abnormal volume are strongly positively correlated, with a correlation of 0.69 at the one-day horizon. In contrast, the text-driven return increment is largely orthogonal to both. The same transcript that leads the model to revise its volatility forecast upward also tends to raise its abnormal-volume forecast, but this movement has little relationship with the return forecast. Earnings-call text therefore generates a coordinated uncertainty-and-attention response without generating coordinated directional beliefs. This is consistent with disagreement-based models in which public signals that investors interpret differently increase uncertainty and trading without necessarily implying a common price direction (Kandel and Pearson, 1995). I further show that the realised-variance increment leads the abnormal-volume increment over horizons of up to three weeks, while the reverse relationship is much weaker. This asymmetry is consistent with gradual information diffusion, where qualitative disclosure initially creates uncertainty that is then resolved through subsequent trading (Hong and Stein, 1999; Vega, 2006).

The paper makes three contributions. First, it provides out-of-sample evidence on the decay of earnings-call text informativeness across returns, realised variance, and abnormal volume within a common forecasting framework. Second, it shows that the main economic value of transcript text lies in forecasting uncertainty and trading activity rather than return prediction.

Third, it documents a mechanism linking qualitative disclosure to market outcomes, showing that text generates a coordinated volatility-volume response, and the volatility component leads subsequent trading activity. The results suggest that earnings-call transcripts shape the process by which markets interpret and absorb information after disclosure events, even when they generate little exploitable directional return predictability.

2. Literature Review

The paper builds on the large literature showing that earnings announcements are central information events in equity markets. Ball and Brown (1968) show that accounting income contains information relevant to security prices, while Beaver (1968) documents abnormal price and trading-volume reactions around earnings releases. Subsequent research shows that this information is not always incorporated immediately. Foster, Olsen and Shevlin (1984) and Bernard and Thomas (1989, 1990) document post-earnings-announcement drift (PEAD), whereby positive earnings surprises are followed by positive abnormal returns and negative surprises by negative abnormal returns. Livnat and Mendenhall (2006) show that the strength of PEAD depends on how earnings surprises are measured, while Chordia, Subrahmanyam and Tong (2014) suggest that some accounting anomalies have weakened over time as liquidity, trading activity, and information processing have improved.

This paper is related to the PEAD literature but differs in emphasis. Rather than estimating in-sample return predictability, it evaluates the out-of-sample predictive content of earnings surprises and earnings-call transcripts across increasing forecast horizons. The objective is not simply to test whether earnings news predicts future returns, but to measure how quickly hard and soft disclosure information is absorbed across different dimensions of market adjustment. The paper also treats returns, realised variance, and abnormal volume as connected outcomes rather than separate exercises. This matters because the joint structure of text-driven predictions helps distinguish between directional price discovery, uncertainty resolution, and disagreement-based trading.

Earnings calls are a natural setting for this question because they supplement financial statements with interpretation, narrative, and forward-looking discussion. Tasker (1998) characterises conference calls as a disclosure mechanism that helps firms bridge information gaps with investors. Frankel, Johnson and Skinner (1999) provide early evidence that calls are associated with voluntary disclosure incentives and market reactions, while Kimbrough (2005) shows that calls can reduce underreaction to earnings announcements. Later work shows that both prepared remarks and analyst Q&A contain useful information (Matsumoto, Pronk and Roelofsen, 2011), and that manager-analyst interactions are themselves informative (Chen, Nagar and Schoenfeld, 2018). This paper follows that literature in treating earnings calls as economically meaningful disclosure events, but shifts the focus from immediate informativeness to the persistence and mechanism of predictive content.

The paper also contributes to the financial textual-analysis literature. Early work relied heavily on dictionary-based measures of tone. Loughran and McDonald (2011) show that general sentiment dictionaries can perform poorly in financial contexts, motivating finance-specific measures. Price et al. (2012) apply tone measures to earnings calls and find that call tone contains incremental information for stock returns. Other studies examine linguistic complexity, obfuscation, deception, and extreme language in corporate communication (Larcker and Zakolyukina, 2012; Bushee, Gow and Taylor, 2018; Bochkay, Hales and Chava, 2020).

More recent work represents text more flexibly using machine-learning methods. Gentzkow, Kelly and Taddy (2019) provide a general framework for using text as data in economics, while Ke, Kelly and Xiu (2019) show that flexible textual representations can contain return-predictive information. Meursault et al. (2023) study textual information in relation to PEAD and show that text can help explain drift after earnings announcements. The present paper is closest to this line of work, but differs in two ways. First, it focuses explicitly on out-of-sample decay: how the incremental value of transcript information changes as the forecast horizon lengthens. Second, it studies the cross-outcome structure of text informativeness, asking whether the same call content jointly predicts uncertainty and trading activity even when it has little directional return content.

This cross-outcome perspective links the paper to theories of disagreement, disclosure processing costs, and gradual information diffusion. Disagreement-based models imply that public signals can generate trading when investors interpret the same information differently, even if the signal does not produce a common directional belief. Harris and Raviv (1993) and Kandel and Pearson (1995) show that heterogeneous interpretation of public information can generate trading volume around announcements. This mechanism is also related to work on disclosure processing costs. Blankespoor, deHaan and Marinovic (2020) argue that public disclosures can remain costly to acquire, monitor, and process, so public information need not be incorporated uniformly across investors. Gradual-diffusion models provide a complementary channel. Hong and Stein (1999) model a market in which information spreads slowly across investors, generating delayed adjustment. Vega (2006) provides evidence that the market response to public news depends on the information environment, consistent with investors processing public information differently across settings. Tetlock (2007) shows that media pessimism predicts market activity, including abnormal trading volume, highlighting the link between qualitative information and trading behaviour. This paper extends these ideas to earnings-call transcripts by asking whether text-driven uncertainty predicts subsequent text-driven trading activity, and whether this lead-lag structure persists across horizons.

Methodologically, the paper follows the shift from dictionary-based textual measures to embedding-based representations. Transformer models generate contextual representations of language (Vaswani et al., 2017), while BERT-style models demonstrate the usefulness of bidirectional pretraining for language-understanding tasks (Devlin et al., 2019). Sentence-BERT adapts these models to produce fixed-length sentence embeddings suitable for downstream prediction tasks (Reimers and Gurevych, 2019). In finance, domain-specific models such as FinBERT have been developed to capture financial vocabulary, tone, and disclosure language more effectively than generic sentiment tools (Araci, 2019; Yang, Uy and Huang, 2020). This paper uses sentence-level embeddings rather than dictionary scores or tone classifications, aggregates them to the call level, and reduces the resulting high-dimensional representation using principal components before estimation.

A further contribution is that the paper studies not only abnormal returns, but also realised variance and abnormal trading volume. These outcomes capture distinct aspects of market adjustment. Returns measure directional price discovery, realised variance captures uncertainty and risk reassessment, and volume reflects attention, disagreement, and portfolio rebalancing. The volatility specification builds on the realised-volatility literature, where intraday returns are used to construct measures of latent return variation (Andersen and Bollerslev, 1998; Andersen, Bollerslev, Diebold and Labys, 2003). HAR-style controls are included following Corsi (2009), ensuring that any textual contribution is incremental to standard volatility persistence. The volume analysis follows research showing that earnings announcements generate trading activity beyond price changes (Beaver, 1968; Bamber, 1987; Bamber, Barron and Stevens, 2011).

Finally, the paper relates to the literature on out-of-sample forecast evaluation. Welch and Goyal (2008) show that many return predictors perform poorly out of sample, while Campbell and Thompson (2008) emphasise the importance of economically meaningful benchmarks in predictive regressions. Gu, Kelly and Xiu (2020) show that machine-learning methods can improve asset-pricing predictions but also highlight the need for rigorous validation. This paper follows that discipline by using chronological train-validation-test splits, out-of-sample R^2, zero-benchmark R^2 for abnormal returns, bootstrap confidence intervals, and Diebold-Mariano tests of predictive accuracy (Diebold and Mariano, 1995). Because cumulative forecast horizons generate overlapping outcomes, Newey-West corrections are used for the loss-differential variance (Newey and West, 1987), with the Harvey, Leybourne and Newbold (1997) small-sample correction. Stationary bootstrap confidence intervals preserve dependence in the ordered test sample (Politis and Romano, 1994), and Holm's (1979) procedure adjusts for multiple comparisons.

The paper synthesises these literatures around a single question: does the qualitative content of earnings calls improve out-of-sample predictions, and if so, through what economic mechanism and for how long? The central finding is that transcript text matters most for uncertainty and trading activity, not for directional return prediction. Text generates coordinated realised-variance and abnormal-volume responses, while remaining largely orthogonal to abnormal returns. Moreover, text-driven realised variance leads subsequent abnormal trading activity with a structured decay over time.

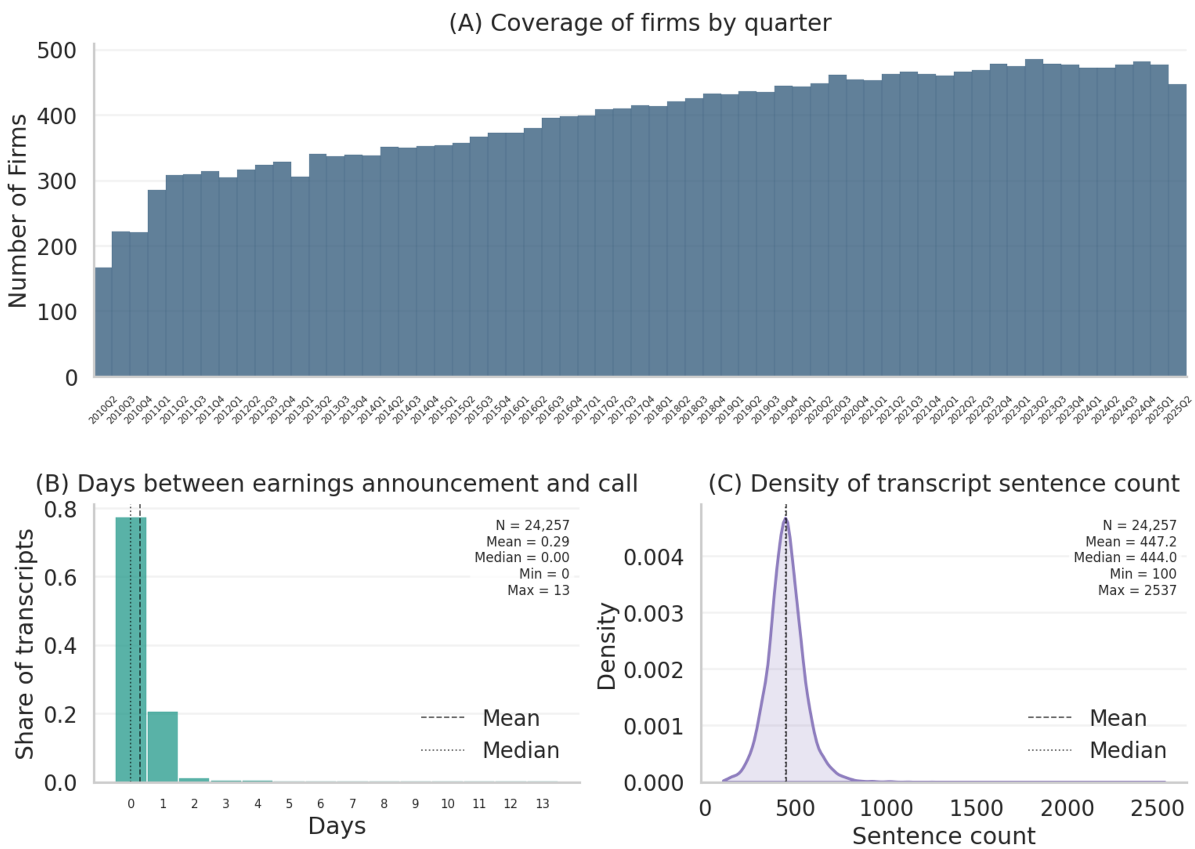
3. Data

3.1. Earnings Call Transcripts

An earnings call is a quarterly conference call held by a publicly listed company, typically following the release of financial statements. During the call, senior executives discuss recent performance, provide forward-looking guidance, and comment on strategic developments, often followed by a Q&A session with analysts.

Earnings-call transcripts for S&P 500 firms from 2010 Q2 to 2025 Q2 were obtained from *Roic.ai* which publishes standardized company transcripts[1]. To avoid survivorship bias, I construct a time-varying panel of S&P 500 constituents using a dynamically compiled list scraped from Wikipedia and maintained on GitHub[2], and retain only firms that are members of the index at the time of each call (and in the days immediately preceding it). The final dataset contains over 24,000 transcripts for 648 unique firms.

Figure 1: Earnings call summary statistics



Note: Panel (A) shows the number of transcripts/firms present in that quarter in the dataset. Panel (B) shows the distribution of firms' number of days between earnings announcement and subsequent earnings call. Panel (C) shows the distribution of sentence count within the transcript dataset.

[1] https://www.roic.ai/.

[2] See repo: https://github.com/fja05680/sp500. Accessed 17-01-2026.

Prior to constructing textual features, each transcript segment is cleaned using a standardized normalization pipeline to ensure consistent inputs across the corpus. The pipeline includes: (i) lowercasing to reduce spurious distinctions due to capitalization; (ii) light formatting normalization (standardizing whitespace and removing transcript artefacts such as repeated separators); and (iii) removal of non-linguistic characters when they do not convey semantic content in this context. The output of this step is a cleaned sequence of text segments that serves as the standardized input for the embedding pipeline described subsequently.

3.1.1. Large Language Models and Embedding Representations

Earnings-call transcripts are represented using **`BAAI/bge-base-en-v1.5`**, a 768-dimensional sentence-embedding model designed for semantic similarity and retrieval tasks. I use an embedding model rather than a dictionary-based tone measure because the objective is not to classify sentiment alone, but to capture a broader representation of the call's semantic content, including discussion of demand conditions, risk, uncertainty, guidance, strategic developments, and analyst interaction. This is useful in earnings calls, where economically relevant information may be expressed through framing, qualification, or narrative structure rather than through a fixed set of keywords[3].

Each cleaned transcript for firm $i$on date t is split into sentences $\{s_1, \dots, s_K\}$. Each sentence is embedded as

$$\psi(s_k) \in \mathbb{R}^{768},$$

producing a sentence-level embedding matrix

$$Z_{i,t}^{\text{sentence}} = \begin{bmatrix} \psi(s_1) \\ \psi(s_2) \\ \vdots \\ \psi(s_K) \end{bmatrix} \in \mathbb{R}^{K \times 768}.$$

I then construct a call-level representation by mean pooling across sentences:

$$e_{i,t} = \frac{1}{K} \sum_{k=1}^{K} \psi(\, s_k) \in \mathbb{R}^{768}.$$

[3] While domain-specific language models such as FinBERT have been developed to capture financial vocabulary, tone, and disclosure language, the objective in this paper is not sentiment classification but continuous semantic representation for downstream forecasting. FinBERT is primarily designed for financial sentiment analysis and is typically fine-tuned to classify text into sentiment categories, whereas **`BAAI/bge-base-en-v1.5`** is designed as a sentence-embedding model for semantic similarity and retrieval tasks. This is relevant because the forecasting exercise uses the geometry of transcript representations, rather than discrete sentiment labels, to predict returns, realised variance, and abnormal volume. BGE-style embedding models are also directly evaluated on broad embedding benchmarks such as the Massive Text Embedding Benchmark, whereas sentiment-tuned financial models are usually evaluated on classification tasks.

This pooled vector provides a compact representation of the transcript's semantic content and serves as the text input for the downstream forecasting models[4]. Because the resulting embedding is high-dimensional and strongly collinear, the forecast design reduces it using principal components estimated only on the training sample.

3.2. Financial data

Intraday 5-minute open, high, low, close and volume (OHLCV) data for S&P 500 constituent firms and the SPY ETF are obtained from Financial Modelling Prep (FMP)[5]. I also obtain earnings-related information from FMP, including earnings announcement dates, reported earnings, and analyst expectations. This data is merged with earnings-call transcript data at the firm-date level.

Using the 5-minute intraday bars, I construct daily measures of returns, realised volatility, and trading volume for each stock and for SPY. These daily series are then used to build forecasting targets and control variables around earnings-related events. The empirical design focuses on post-event outcomes over multiple horizons. In the main specification, the information set is restricted to data available strictly before the event date, while forecast windows begin on the next trading day after the event.

3.2.1. Abnormal returns

Let daily log returns be

$$R_{i,t} = \log\left(P_{i,t}\right) - \log\left(P_{i,t-1}\right), \qquad R_t^{mkt} \text{ for SPY.}$$

Estimate a firm-specific market beta using pre-event data:

$$\beta_i = \frac{\mathrm{Cov}(R_{i,t}, R_t^{mkt})}{\mathrm{Var}(R_t^{mkt})}.$$

Abnormal returns are

$$AR_{i,t} = R_{i,t} - \beta_i R_t^{mkt}.$$

[4] Mean pooling across all sentence embeddings is a deliberate design choice motivated by parsimony and out-of-sample discipline. Alternative aggregation strategies, such as pooling prepared remarks and Q&A separately, applying attention-weighted pooling, or preserving only high-salience sentences, may capture finer structure within the call but also introduce additional modelling choices. Prior work shows that prepared remarks and analyst Q&A contain distinct informational content (Matsumoto, Pronk and Roelofsen, 2011), so a more granular decomposition is a natural extension. In the present setting, I use the document-level pooled embedding as a low-dimensional summary of the full call, leaving a more granular analysis of call structure to future work.
[5] https://site.financialmodelingprep.com/.

Cumulative abnormal return over horizon H is

$$CAR_{i,t}^{(H)} = \sum_{h=1}^{H} AR_{i,t+h}.$$

Appendix Figure A1 reports the distribution of CAR across horizons.

3.2.2. Realised Variance

For firm i, trading day t, and intraday interval $j = 1, \dots, J_t$, let the 5-minute log return be

$$r_{i,t,j} = 100\left[\log\left(P_{i,t,j}\right) - \log\left(P_{i,t,j-1}\right)\right],$$

where $P_{i,t,j}$ denotes the transaction price at interval j. Multiplying by 100 expresses returns in percentage-point units. Daily realised volatility is then defined as the sum of squared 5-minute returns within the day,

$$RV_{i,t} = \sum_{j=1}^{J_t} r_{i,t,j}^2.$$

For an event anchored at day t, the H-day realised-variance outcome is measured over the next H trading days:

$$RV_{i,t}^{(H)} = \sum_{h=0}^{H-1} RV_{i,t+h}.$$

In the empirical implementation, the volatility target is stored as

$$\log\left(1+RV_{i,t}^{(H)}\right).$$

Thus, for an event occurring on day t, the H-day volatility outcome is the log-transformed cumulative realised volatility from the first trading day after the event through the next H trading days. In addition to these targets, I construct HAR-style volatility controls using pre-event information only: the log of 1-day, 5-day and 21-day average realised volatility for both the firm and SPY. Appendix Figure A2 reports the distribution of $\log\left(1+RV_{i,t}\right)$ across horizons.

3.2.3. Abnormal trading volume

Let $V_{i,t}$ denote total daily trading volume for firm i on day t, obtained by summing intraday 5-minute volume across the trading day. For an H-day horizon, cumulative post-event volume is

$$VOL_{i,t}^{(H)} = \sum_{h=0}^{H-1} V_{i,t+h}.$$

To scale this quantity, I compare it with expected volume based on the firm's pre-event average daily volume over 21-days. If $\bar{V}_{i,t-1}$ denotes the average daily volume over a pre-event estimation window, expected cumulative volume over horizon H is given by $H\bar{V}_{i,t-1}$. Abnormal volume is then measured as

$$ABVOL_{i,t}^{(H)} = \log\left(1 + \frac{VOL_{i,t}^{(H)}}{H\bar{V}_{i,t-1}}\right).$$

Accordingly, the abnormal-volume outcome captures the extent to which post-event trading activity exceeds typical pre-event trading volume over the same horizon. As with realised volatility, the controls are based only on pre-event information. The distribution of abnormal volume across horizons in the analysis sample is documented in appendix Figure A3.

3.2.4. Earnings surprises

Earnings surprises capture the extent to which reported earnings deviate from market expectations and are a key driver of post-announcement price reactions. The standardised earnings surprise (SUE) is defined as

$$SUE_{i,q} = \frac{EPS_{i,q} - \mathbb{E}[EPS_{i,q}]}{\sigma(EPS_i)}$$

where $EPS_{i,q}$ is the reported earnings per share of firm i in quarter q, $\mathbb{E}[EPS_{i,q}]$ is the consensus analyst forecast immediately prior to the announcement, and $\sigma(EPS_i)$ is the historical standard deviation of forecast errors for firm i[6]. This scaling makes the surprise comparable across firms and time. I also record the number of days between firm i's earnings announcement and their earnings call.

Table 1 reports summary statistics for the main event-level covariates used in the forecasting design. Appendix Figures A1–A6 provide the corresponding distributions.

[6] $\sigma(EPS_i)$ is calculated using firm i's earnings information from the previous 8 quarters, with a minimum of 4 consecutive quarters required for the measure to be computed. This adjustment accounts for the fact that some firms historically exhibit greater earnings volatility than others. As a result, raw earnings surprises should not be compared naively across firms, since the same EPS surprise may represent a very different degree of unexpectedness depending on the firm's historical variability.

Table 1: Summary statistics - analysis sample

Variable	*N*	*Mean*	*Median*	*Std*	*p10*	*p90*	*Min*	*Max*
SUE	24366	0.907	0.645	1.607	-0.555	2.787	-3.767	7.375
|SUE|	24366	1.269	0.862	1.339	0.076	2.955	0.000	7.375
Days between EA and call	24366	0.289	0.000	0.754	0.000	1.000	0.000	13.000
Pre-event market beta (252-day)	24366	0.995	0.983	0.388	0.498	1.498	0.129	2.078
Pre-event avg. daily volume (millions)	24366	4.644	2.055	8.131	0.595	9.887	0.198	56.378
Transcript sentence count	24366	447	444	109	323	569	100	2537

Note: Summary statistics for key covariates in the analysis sample. S&P 500 constituents, 2010Q2-2025Q2. Sample restricted to transcripts with at least 100 sentences. Listwise deletion applied on all baseline and earnings covariates across all outcomes.

4. Forecast Design

The empirical objective is to measure how long earnings-call information retains incremental predictive content once one conditions on publicly observable quantitative information available at the time of the call. Let i index earnings-call events, h denote the forecast horizon in trading days, and $j \in \{\text{abnormal returns, realised variance, abnormal volume}\}$ index the outcome family. The design traces the horizon over which information fixed at the call continues to predict future market outcomes out of sample.

The analysis is organised around each firm's earning-call. This anchor is defined relative to the start of the earnings call, after the qualitative information conveyed through management discussion and analyst interaction has started to become available. When the call clock time is unavailable, the anchor is set to the next regular-session opening bar. The call anchor is therefore the point at which both the hard earnings information and the qualitative transcript information can be treated as available to the forecasting model.

For each horizon h, the outcome is defined as the cumulative quantity over the h-day trading window following the call anchor, and the forecast is constructed using only information available at the call. The design therefore asks how far ahead the information set observed at the call continues to predict future returns, volatility, or trading activity.

Let $Y_{i,h}^{(j)}$ denote the outcome for event i, horizon h, and outcome family j. Five nested specifications are estimated. The baseline model uses only non-text controls:

$$Y_{i,h}^{(j)} = f_h^{(j)}(X_i^{(0)}) + \varepsilon_{i,h}^{(j)}.$$

The second specification augments the baseline with hard earnings information:

$$Y_{i,h}^{(j)} = f_h^{(j)}(X_i^{(0)}, X_i^{(E)}) + \varepsilon_{i,h}^{(j)}.$$

The third adds transcript-based text features while omitting earnings variables:

$$Y_{i,h}^{(j)} = f_h^{(j)}(X_i^{(0)}, Z_i) + \varepsilon_{i,h}^{(j)}.$$

The fourth is the fully augmented model, combining baseline controls, earnings variables, and transcript features:

$$Y_{i,h}^{(j)} = f_h^{(j)}(X_i^{(0)}, X_i^{(E)}, Z_i) + \varepsilon_{i,h}^{(j)}.$$

Finally, a text-only specification is estimated:

$$Y_{i,h}^{(j)} = f_h^{(j)}(Z_i) + \varepsilon_{i,h}^{(j)}.$$

This specification tests whether transcript content carries predictive signal before conditioning on the structured financial environment. The comparison between the earnings-plus-text model and the earnings-only model is the most direct test of whether transcript information adds predictive content beyond the hard earnings information available by the time of the call.

The baseline control block $X_i^{(0)}$ varies by outcome family. For realised variance, it includes lagged firm-specific realised variance measured over one-, five-, and twenty-one-day windows, together with analogous SPY-based market volatility controls. For abnormal volume, it includes pre-event average daily trading volume over a twenty-one-day window. For abnormal returns, it includes the firm's pre-event market beta.

The hard-information block $X_i^{(E)}$ contains standardized unexpected earnings, its absolute value, and the number of days between the earnings release and the call[7]. This block captures both the signed magnitude of the earnings news and the timing of the call relative to the announcement.

Transcript information enters through a document-level embedding representation constructed from the full earnings-call transcript. Let $e_i \in \mathbb{R}^{768}$ denote the mean-pooled transcript embedding for event i. Because these embedding vectors are high-dimensional and strongly collinear, the text block is introduced through principal components of the standardised embedding space. If P_k denotes the matrix of the first k principal components, the reduced transcript representation is

$$Z_i^{(k)} = P_k^{\top} \tilde{e}_i,$$

where $\tilde{e}_i$ is the standardised embedding. The number of retained components is selected from $k \in \{10,20,50,100\}$using validation-sample performance. This choice is made separately by outcome, horizon, specification, and model family for the call-anchor text specifications. During model selection, the PCA transformation is fitted on the training sample and applied to the validation sample. After tuning, the final text transformation and forecasting model are estimated on the combined training and validation sample before evaluation on the held-out test sample.

The estimation method varies across outcome families. For abnormal returns, the primary forecasting model is ridge regression with an unrestricted intercept. This reflects the weak and noisy nature of return predictability and the tendency of ridge regularisation to deliver more stable out-of-sample forecasts in this setting. At each horizon h, the estimator solves

[7] If day of announcement coincides with day of call, days difference is equal to 0.

$$\hat{\beta}_h(\lambda) = \arg\min_{\beta} \sum_{i \in T_{\text{train}}} \left(Y_{i,h} - \alpha - W_i^{\top}\beta\right)^2 + \lambda \|\beta\|_2^2,$$

where W_i is the feature vector implied by the relevant specification and λ is selected using the validation sample.

For realised variance and abnormal volume, the primary estimator is LightGBM, an implementation of gradient-boosted regression trees. At horizon h, the fitted predictor can be written as

$$\hat{g}(x) = \sum_{m=1}^{M} \eta\, T_m(x),$$

where $T_m(\cdot)$ denotes a regression tree, η is the learning rate, and M is the number of boosting iterations. Hyperparameters are selected on the validation sample, which is also used for early stopping. As a robustness exercise, OLS, ridge, and LightGBM are estimated in parallel, comparing the baseline specification with the fully augmented baseline-plus-earnings-plus-text specification[8].

All forecasts are evaluated under a strictly chronological train-validation-test design. Unique event dates are ordered through time and partitioned into training, validation, and test periods using fixed shares of 60, 15, and 25 percent. Model choices are made using the training and validation samples only. After selecting the relevant hyperparameters and, where applicable, the number of text principal components, the selected model is refit on the combined training and validation sample and evaluated once on the held-out test sample. This ensures that every reported performance measure is genuinely out of sample.

The estimation sample is restricted in several steps. First, transcripts with fewer than 100 sentences are excluded, as these observations may correspond to incomplete or partially captured calls[9]. Second, outcome variables and continuous covariates are winsorised at the 1st and 99th percentiles to limit the influence of extreme observations on model estimation. Winsorisation bounds are estimated on the training sample only and then applied to the validation and test samples, ensuring that no information from future periods enters the treatment of extreme values. This is particularly relevant for realised variance and abnormal volume, which are strongly right-skewed and can exhibit large spikes around earnings events. Finally, observations are dropped listwise within each outcome-horizon cell. Any event with a missing outcome or a missing value in a covariate entering that cell is excluded from the

[8] A comparison of model performance is presented in Appendix A.

[9] This filter is necessitated by an artefact of the Roic.ai data source. Downloaded transcripts cannot be verified as complete, with some entries appearing to represent only a fragment of the actual call, potentially due to partial API coverage, incomplete parsing, or truncated uploads. There is no reliable indicator in the raw data distinguishing a genuinely short call from an incomplete one. A threshold of 100 sentences is applied as a practical lower bound, since a complete earnings would typically run to several hundred sentences for an S&P 500 firm. Transcripts falling below this threshold are therefore more likely to be data quality issues than genuine short calls, and their inclusion could introduce measurement error in the embedding representations. The 100-sentence cutoff retains the vast majority of observations while removing entries that would produce unreliable document-level embeddings due to insufficient text content.

estimation and evaluation sample. The complete-case restriction is applied after winsorisation and is held fixed across all nested specifications within the same outcome-horizon cell, so that performance comparisons are based on identical samples.

Forecast performance is evaluated differently across outcome families. For realised variance and abnormal volume, the primary metric is conventional out-of-sample R^2:

$$R_h^2 = 1 - \frac{\sum_{i \in T_{\text{test}}} \left(Y_{i,h} - \hat{Y}_{i,h}\right)^2}{\sum_{i \in T_{\text{test}}} \left(Y_{i,h} - \bar{Y}_{\text{test}}\right)^2}.$$

For abnormal returns, the headline metric is zero-benchmark out-of-sample R^2:

$$R_{\text{zero},h}^2 = 1 - \frac{\sum_{i \in T_{\text{test}}} \left(Y_{i,h} - \hat{Y}_{i,h}\right)^2}{\sum_{i \in T_{\text{test}}} Y_{i,h}^2}.$$

This is appropriate because the economically meaningful benchmark for abnormal returns is zero abnormal performance rather than the realised test-sample mean (Campbell and Thompson, 2008).

Uncertainty around standalone R^2 values and pairwise differences in R^2 is quantified using the stationary bootstrap. The bootstrap resamples blocks of geometrically distributed length, preserving serial dependence in the ordered test sample without imposing a fixed block size. In the implementation, the average block length is set to $\max(5, h)$, allowing the dependence adjustment to increase with the forecast horizon. For each pairwise comparison, the stationary bootstrap is also used to compute p-values for ΔR^2.

Formal model comparison is based on pairwise tests of incremental predictive ability. The main comparisons are between the earnings-augmented and baseline models, between the text-augmented and baseline models, between the earnings-plus-text and baseline models, between the earnings-plus-text and earnings-only models, and between the text-only and baseline models. The comparison between the earnings-plus-text and earnings-only models is the most direct test of whether transcript information adds predictive content beyond the hard earnings release.

Diebold-Mariano statistics are computed using squared forecast-error losses on the test sample. Under the null of equal predictive accuracy, the expected loss differential is zero. The variance of the loss differential is estimated using a Newey-West heteroskedasticity- and autocorrelation-consistent estimator with lag length $h - 1$, which accounts for the mechanical overlap induced by cumulative h-day outcome windows. The Harvey-Leybourne-Newbold small-sample correction is applied to the resulting statistic.

Because multiple tests are conducted across horizons and nested comparisons, p-values are adjusted using the Holm-Bonferroni step-down procedure. The correction is applied separately within each family defined by analysis type, anchor, outcome family, and model family. Thus, for the primary call-anchor results, the adjustment controls across all horizons and all ΔR^2

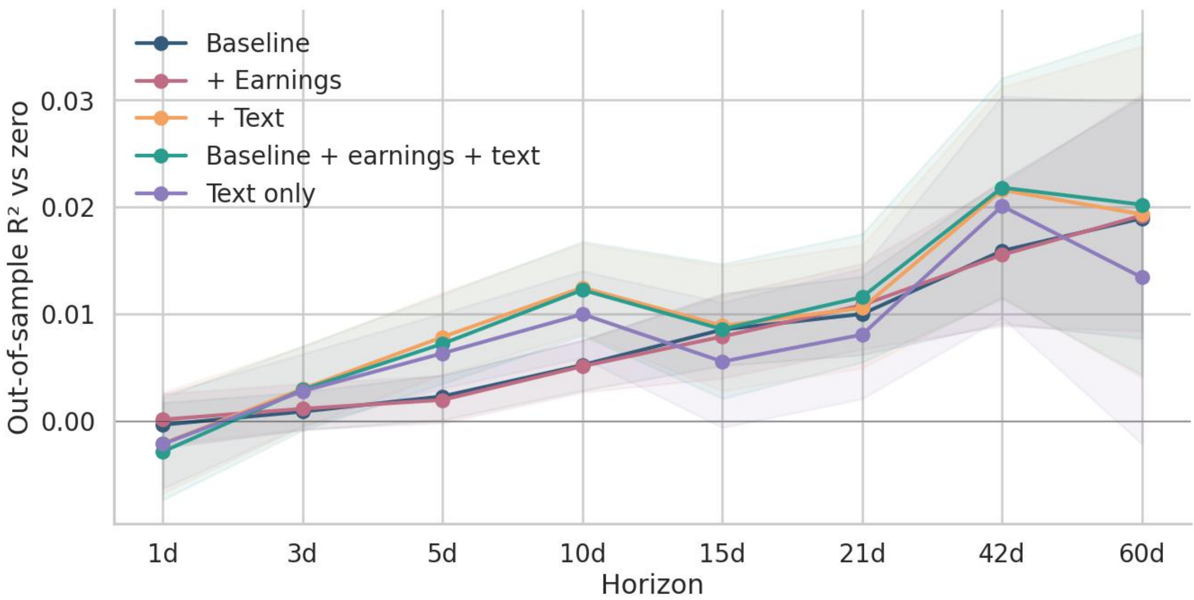
comparisons for a given outcome and model family. The same correction is applied separately to the stationary-bootstrap ΔR^2 p-values and to the Newey-West Diebold-Mariano p-values.

5. Results

5.1. Abnormal Returns

Figure 2 shows out-of-sample performance for abnormal-return forecasts across horizons. Predictability increases modestly with the forecast horizon, consistent with the gradual averaging out of idiosyncratic noise over longer cumulative windows. Specifications that incorporate transcript text, either alone or in combination with earnings variables, outperform the baseline at medium horizons. By contrast, adding earnings variables alone provides little incremental predictive power at any horizon.

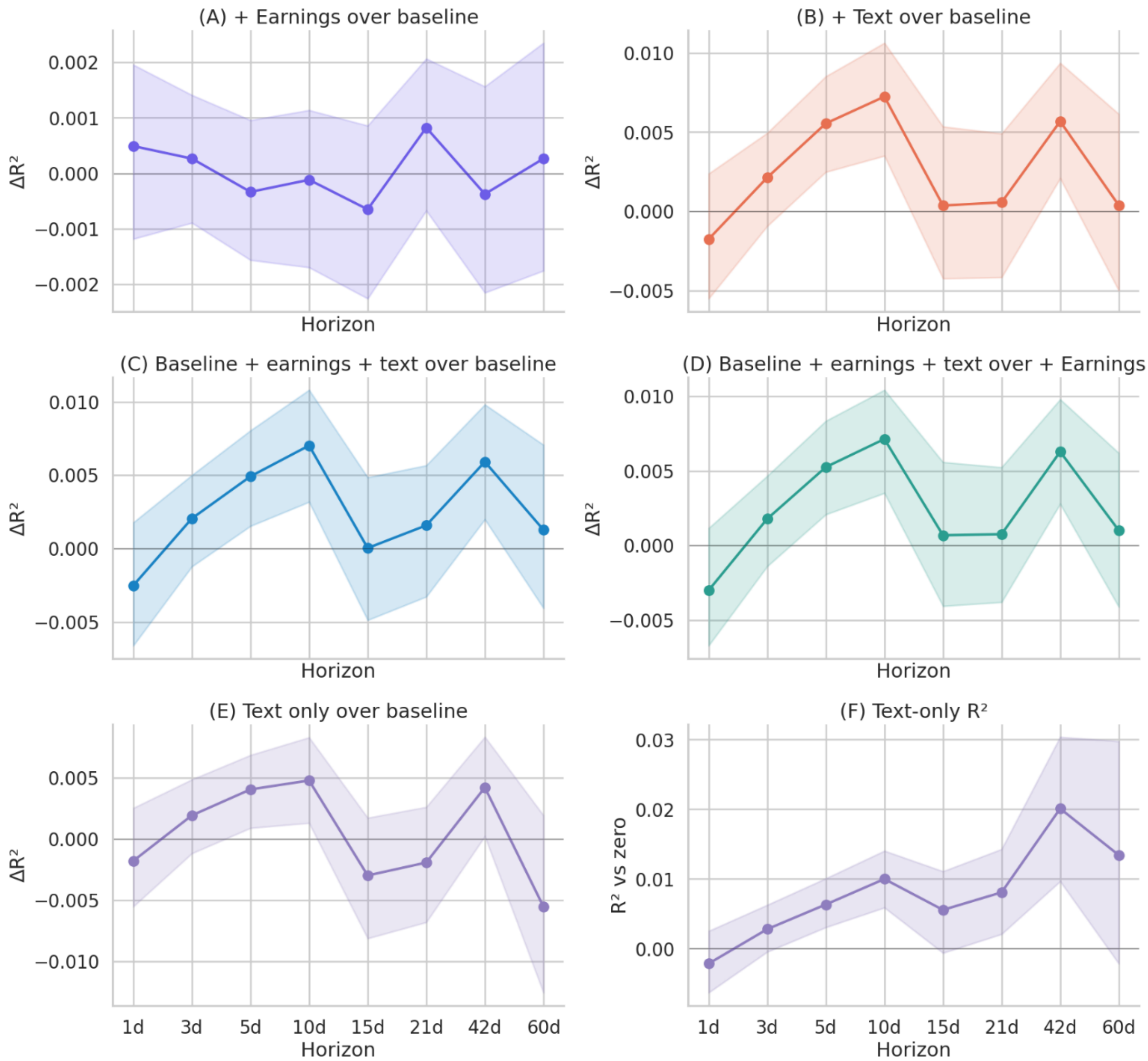
Figure 2: Abnormal returns



Note: The figure reports out-of-sample R^2 vs zero for abnormal-return forecasts across horizons. Confidence intervals are Holm-Bonferroni-adjusted 95% intervals calculated using the stationary bootstrap.

Figure 3 isolates these patterns by reporting pairwise R^2 differences. Adding transcript text yields statistically significant improvements over the baseline at approximately the five- to ten-day horizons. Earnings variables alone do not improve performance, and combining earnings variables with text adds little beyond the text specification. The text-only specification also performs better than a zero benchmark at several horizons, suggesting that earnings-call transcripts contain some independent predictive content not fully captured by conventional controls.

Figure 3: Abnormal returns R^2 comparison



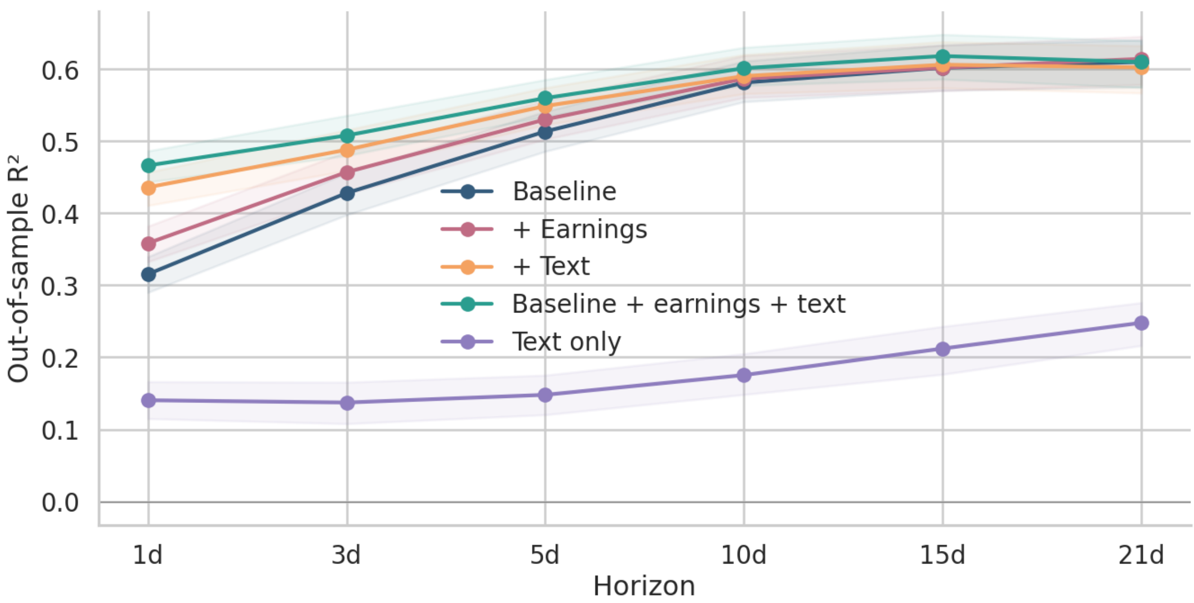
Note: Panels (A)–(E) report differences in out-of-sample R^2 between the baseline specification and alternative specifications. Confidence intervals are Holm-Bonferroni-adjusted 95% intervals for ΔR^2, calculated using the stationary bootstrap. Panel (F) reports standalone out-of-sample performance for the text-only specification with analogous confidence intervals.

The RMSE comparisons in Appendix Table A3 confirm that these return-predictive gains are economically small. Text-related reductions in RMSE are concentrated at the 5- and 10-day horizons and are at most around 0.35%. Thus, while transcript text contains statistically detectable return information at medium horizons, its economic magnitude is modest[10].

5.2. Realised Variance

Figures 4 and 5 present the realised-variance results. In contrast to abnormal returns, transcript text provides substantial short-horizon improvements in predictive performance. Specifications that include text, especially the specification combining earnings variables and text, outperform both the baseline and earnings-only models at horizons up to approximately five trading days.

Figure 4: Realised variance



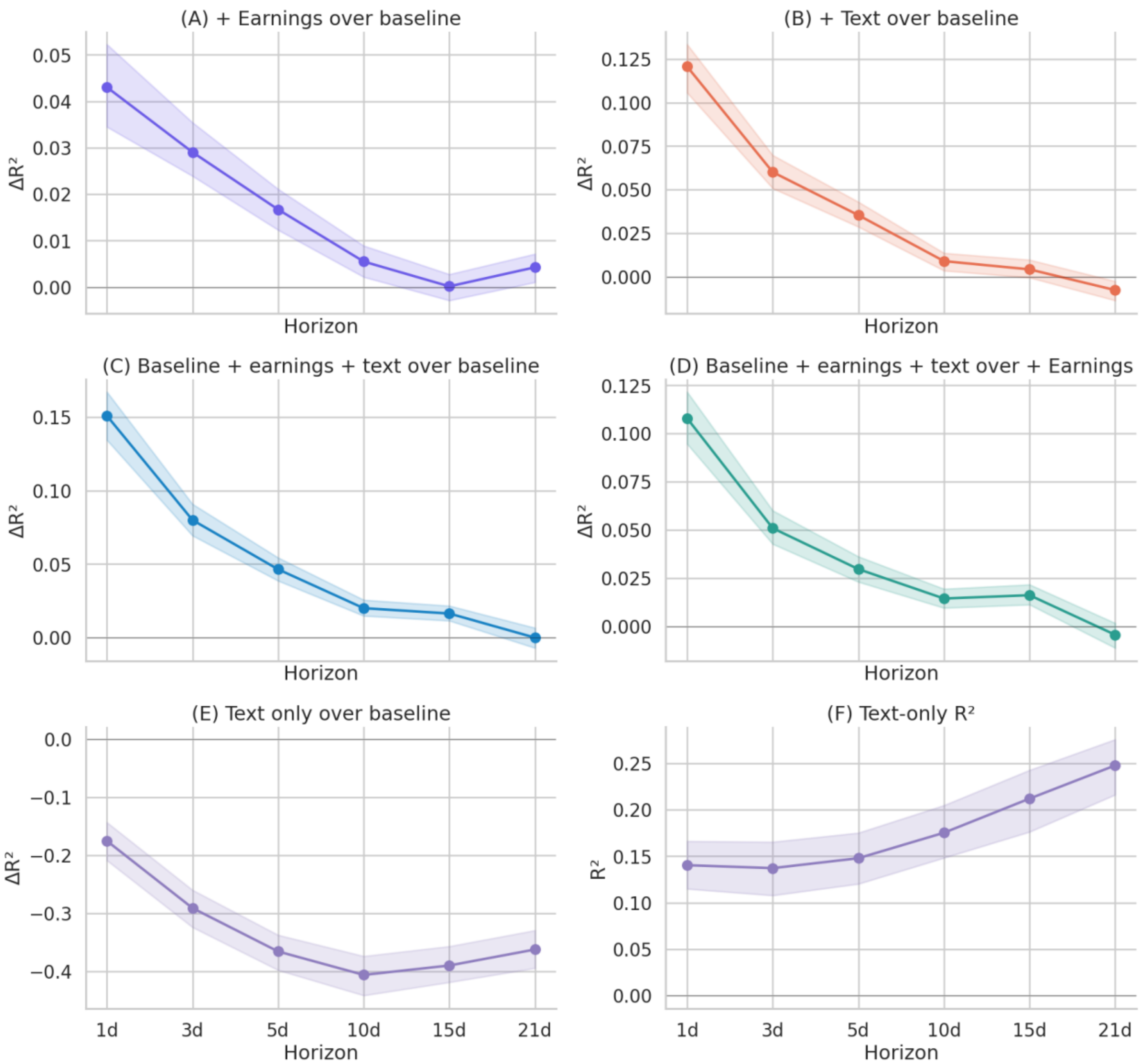
Note: The figure reports out-of-sample R^2 for realised-variance forecasts across horizons. Confidence intervals are Holm-Bonferroni-adjusted 95% intervals calculated using the stationary bootstrap.

Figure 5 reports the corresponding R^2 comparisons. The short-horizon gains from text are economically meaningful, with R^2 improvements of approximately 0.10–0.15 relative to the

[10] While the RMSE improvements for abnormal returns are modest, Appendix B shows that they correspond to some cross-sectional ranking power at medium horizons. A simple equal-weighted long-short quintile portfolio ranked on the text-augmented CAR prediction yields annualised Sharpe ratios of 0.73 at the 5-day horizon and 0.70 at the 10-day horizon, compared with 0.45 and 0.42 for the same strategy ranked on the earnings-only prediction. These are equal-weighted paper returns with no transaction cost optimisation, liquidity adjustment, short-sale constraints, leverage, or overlapping-position treatment. Even modest execution costs could substantially reduce the reported Sharpe differentials. The exercise is therefore best interpreted as evidence of gradual information absorption and limits to arbitrage, not as evidence of directly implementable abnormal returns. Full details are reported in Appendix B.

baseline at the shortest horizons. However, the advantage declines beyond ten trading days and is largely exhausted by 21 days. This pattern is consistent with the strongly autoregressive nature of realised variance. Short-run volatility dynamics are affected by new information released during the call, but longer-horizon forecasts are increasingly dominated by volatility persistence captured by HAR-style controls. Text-only specifications perform poorly relative to specifications that include autoregressive volatility controls, particularly at longer horizons, confirming the importance of conditioning on the standard volatility environment.

Figure 5: Realised variance R^2 comparison



Note: Panels (A)–(E) report differences in out-of-sample R^2 between the baseline specification and alternative specifications. Confidence intervals are Holm-Bonferroni-adjusted 95% intervals for ΔR^2, calculated using the stationary bootstrap. Panel (F) reports standalone out-of-sample performance for the text-only specification with analogous confidence intervals.

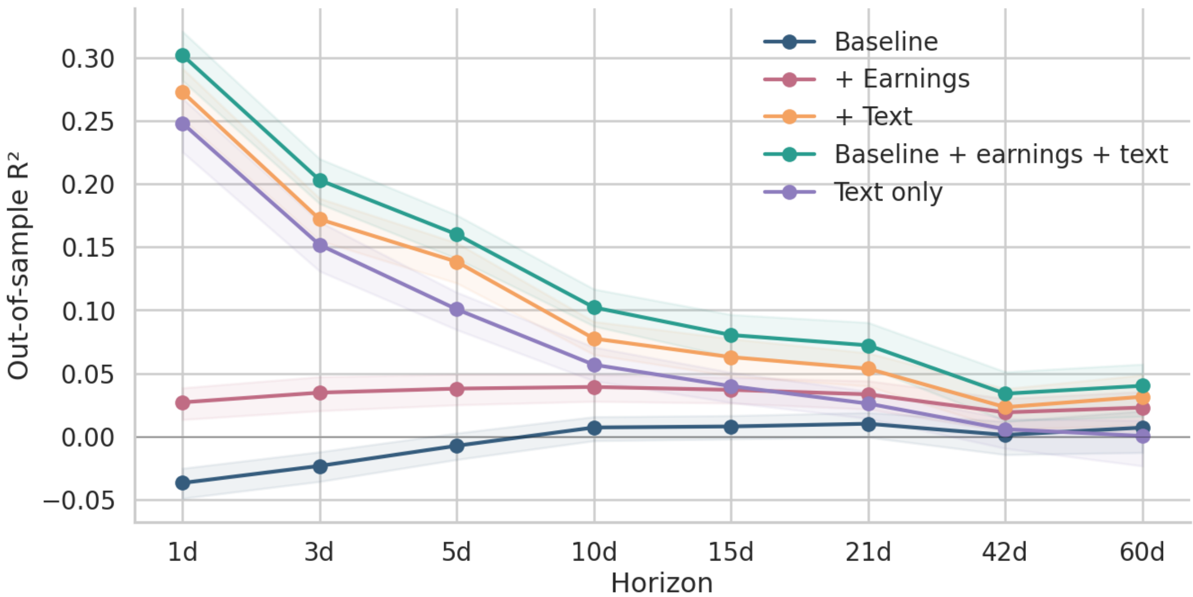
The RMSE results in Appendix Table A4 reinforce this interpretation. At short horizons, text-based models yield large forecast improvements. Relative to the baseline, the fully augmented earnings-plus-text model reduces RMSE by 11.7% at the one-day horizon and 7.3% at the three-day horizon. Relative to the earnings-only model, adding text reduces RMSE by 8.8% and 4.8% at the same horizons.

Overall, the evidence suggests that earnings-call transcripts contain information about short-term uncertainty and risk reassessment immediately after the call, but that this information decays relatively quickly.

5.3. Abnormal Volume

Figures 6 and 7 present results for abnormal trading volume. This is the outcome for which transcript text has the strongest predictive content. Baseline and earnings-only specifications perform relatively poorly, while specifications incorporating text produce large improvements in out-of-sample R^2.

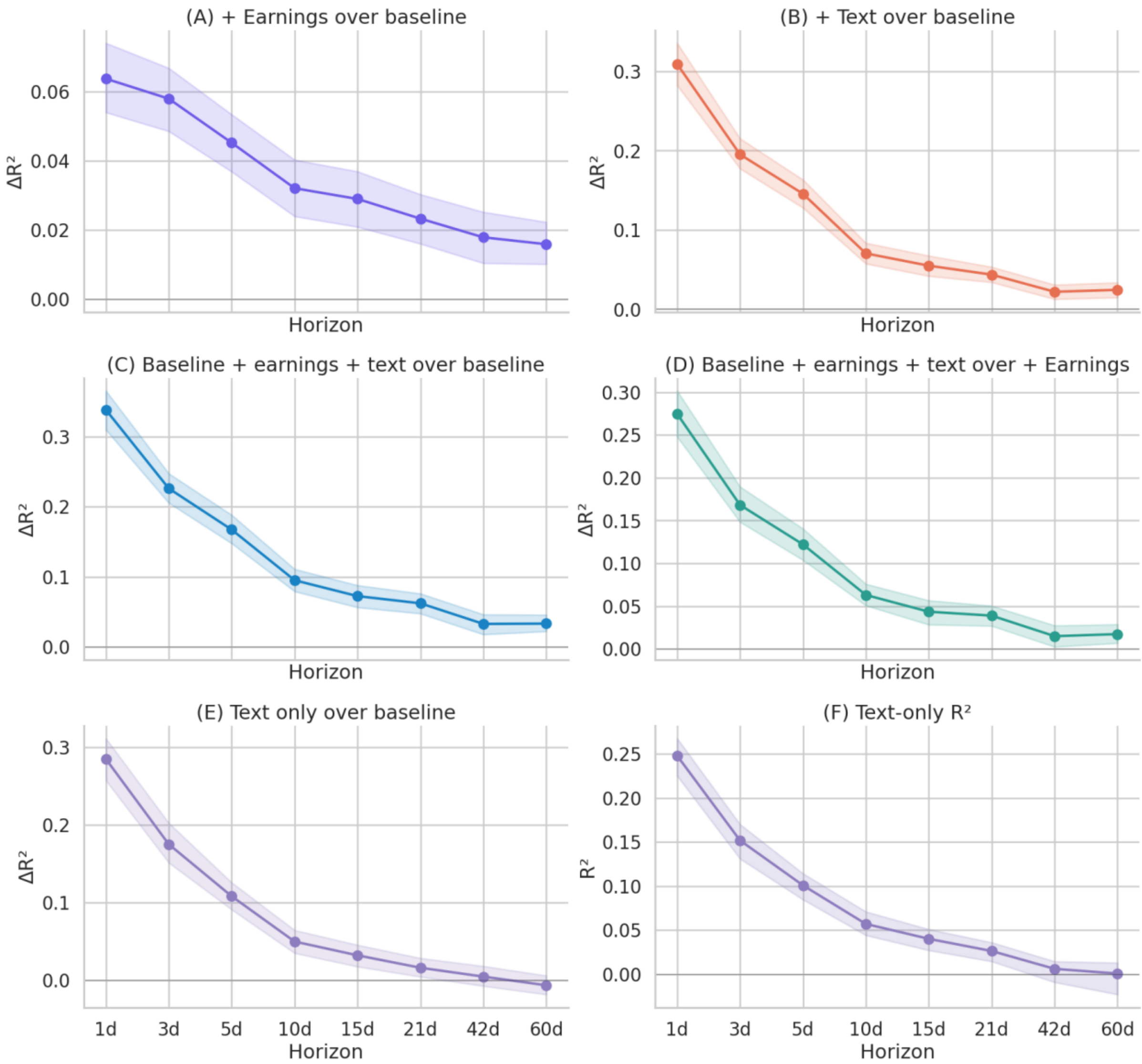
Figure 6: Abnormal volume



Note: The figure reports out-of-sample R^2 for abnormal-volume forecasts across horizons. Confidence intervals are Holm-Bonferroni-adjusted 95% intervals calculated using the stationary bootstrap

At short horizons, the gains are substantial. Including both earnings variables and text increases R^2 by approximately 0.30 at the one-day horizon, while text alone produces an increase of approximately 0.20. Although the magnitude declines over time, the improvements remain positive and economically meaningful even at longer horizons. This pattern is consistent with earnings calls shaping investor attention, disagreement, and trading behaviour. Unlike returns, which appear to incorporate directional information relatively quickly, trading volume reflects the process of information diffusion and heterogeneous belief updating, generating more persistent effects.

Figure 7: Abnormal volume R^2 comparison



Note: Panels (A)–(E) report differences in out-of-sample R^2 between the baseline specification and alternative specifications. Confidence intervals are Holm-Bonferroni-adjusted 95% intervals for ΔR^2, calculated using the stationary bootstrap. Panel (F) reports standalone out-of-sample performance for the text-only specification with analogous confidence intervals.

The RMSE comparisons in Appendix Table A5 confirm these findings. Text-based specifications deliver large improvements at short horizons, with one-day RMSE reductions of approximately 15–18%, depending on the comparison. The magnitude declines gradually over time, but remains positive for the fully augmented model even at the 60-day horizon.

The results indicate that transcript text improves out-of-sample forecasting performance most strongly for realised variance and abnormal volume, and only weakly for abnormal returns. The next section evaluates whether these gains are specific to matched firm-event transcript content, rather than being driven by common quarter-level conditions or persistent firm-level communication style[11].

5.4. Placebo tests

I consider two alternative explanations for the text-based forecasting gains. First, the text increment may capture quarter-level macroeconomic or market conditions that are common to all firms reporting in the same calendar quarter. If so, the relevant information would not be specific to the matched firm-call pair, and shuffling text increments across firms within the same quarter should preserve much of the predictive gain. Second, the text increment may capture persistent firm-level communication style rather than time-varying event-specific disclosure. If so, shuffling text increments across quarters within the same firm should leave much of the forecasting performance intact.

I test these alternatives using two permutation placebo designs. In both cases, the object being permuted is the event-level text increment,

$$\Delta(h) = \hat{y}^{EA+Text}(h) - \hat{y}^{EA}(h),$$

which measures the marginal contribution of transcript text relative to the earnings-only model. In Placebo A, the text increment is randomly reassigned across firms within the same calendar quarter. This preserves common quarter-level information while breaking the match between a firm and its own call content. In Placebo B, the text increment is randomly reassigned across quarters within the same firm. This preserves persistent firm-level communication features while breaking the match between the transcript and the specific earnings event.

In both designs, the earnings-only prediction is held fixed, and the permuted text increment is added back to it. Any loss of incremental R^2 therefore reflects the destruction of the matched transcript-event link. The test statistic is the incremental R^2 of the text-augmented prediction relative to the earnings-only prediction under permutation. Placebo distributions are constructed using 500 permutation draws, and p-values are computed as the fraction of draws with incremental R^2 greater than or equal to the true incremental R^2.

Appendix Figures A10-A12 report the full placebo distributions. For realised variance and abnormal volume, both placebo designs are rejected at all horizons, with permutation p-values equal to 0.000. Shuffling text increments across firms within the same quarter or across quarters within the same firm eliminates the incremental forecasting gain. This indicates that the predictive content for realised variance and abnormal volume is specific to the matched

[11] To also assess whether these results are concentrated in specific market conditions, I split the test sample by the level of the VIX index on the call date (above vs below the test-sample median of 18.5) and by the chronological midpoint of the test period (June 2021–May 2023 vs June 2023–May 2025). The text-driven ΔR^2 for abnormal volume is positive in every cell across both splits. For realised variance the short-horizon gains are present in both VIX regimes and both calendar halves, with the 21-day deterioration appearing in both. For abnormal returns the medium-horizon signal is visible in both environments. The results are therefore not concentrated in any particular market regime or sub-episode of the evaluation window. Full results are reported in Appendix C.

transcript and event. It is not simply a quarter-level macro or market condition, nor is it a persistent firm-level communication style.

For abnormal returns, Placebo A is rejected at all horizons, with p-values below 0.01, indicating that the return-related text signal is not driven by common quarter-level conditions. Placebo B is rejected at the five- and ten-day horizons, where the main return signal is strongest, with p-values of 0.008 and below 0.001, respectively. At the one- and fifteen-day horizons, however, the same-firm placebo is not rejected at conventional levels. This pattern is consistent with the weaker return results, as the signal does appear event specific at horizons where transcript text materially improves return forecasts (5- and 10-days).

5.5. Cross-outcome text informativeness

The preceding sections show that transcript text improves forecasts of realised variance and abnormal volume but provides only weak gains for abnormal returns. These outcome-specific results establish where textual information is most predictive, but they do not reveal whether the effects are connected. A natural question is whether the same earnings call generates a coordinated market response across uncertainty and trading activity, or whether the predictive gains for realised variance and abnormal volume reflect separate signals.

The disagreement-based interpretation implies a specific cross-outcome structure. If qualitative disclosure creates heterogeneous interpretation, the text signal should generate correlated responses in uncertainty and trading activity, since both arise from ambiguity about firm value. However, the same text signal need not generate correlated return forecasts, because disagreement may raise volatility and volume without producing a common directional belief. I test this by examining the event-level text increment for each outcome and measuring the pairwise correlations across outcomes.

For outcome o and horizon h, define the event-level text increment as

$$\Delta_o(h) = \hat{y}_o^{EA+Text}(h) - \hat{y}_o^{EA}(h).$$

This measures how much the transcript changes the h-day-ahead forecast beyond what is already implied by the earnings variables. The object of interest is t the marginal contribution of qualitative call content relative to the hard earnings information. Because all forecast windows begin at the call, the correlations describe the cross-outcome structure of transcript-driven information absorption after the call.

Table 2 presents the cross-outcome correlation structure. The left-hand columns report pairwise correlations among the text increments for abnormal returns, realised variance, and abnormal volume across horizons. The right-hand columns report the same correlations for the earnings-only predictions, which provide a hard-information benchmark.

The results reveal that the realised-variance and abnormal-volume text increments are strongly correlated at short horizons. Their correlation is 0.69 at the one-day horizon and declines gradually to 0.11 by 21 days. By contrast, the corresponding correlation in the earnings-only predictions is only 0.14 at one day and close to zero thereafter. The co-movement between volatility and trading activity is therefore not a generic feature of earnings events, but is specific to the information contained in the earnings-call transcript.

At the same time, the abnormal-return text increment is largely orthogonal to both realised variance and abnormal volume. Across horizons, the CAR–RV and CAR–VOL text-increment correlations are small and generally negative, ranging approximately from -0.23 to 0.07. This differs from the earnings-only benchmark, where the return component is more strongly negatively correlated with realised variance. The transcript signal therefore does not appear to generate coordinated directional beliefs.

Table 2: Cross-outcome post-event informativeness

	Text increment correlations			**Earnings predictions correlations**		
Horizon	*CAR-RV*	*CAR-VOL*	*RV-VOL*	*CAR-RV*	*CAR-VOL*	*RV-VOL*
1d	-0.172	-0.217	0.691	-0.481	-0.100	0.143
3d	-0.234	-0.230	0.513	-0.437	0.102	0.040
5d	-0.171	-0.102	0.386	-0.441	0.120	0.002
10d	-0.189	-0.037	0.254	-0.382	0.101	-0.044
15d	-0.073	-0.025	0.201	-0.421	0.061	-0.058
21d	-0.101	0.066	0.111	-0.432	-0.007	-0.060

Note: The table reports pairwise Pearson correlations among event-level text increments, defined as $EA + Text$predictions minus EA-only predictions, for abnormal returns (CAR), realised variance (RV), and abnormal volume (VOL). The right-hand columns report the corresponding pairwise correlations among EA-only predictions.

The interpretation is that earnings-call text generates a coordinated uncertainty-and-attention response, rather than a coordinated directional return response. The same transcript that leads the model to revise its volatility forecast upward also tends to raise its abnormal-volume forecast. This is consistent with qualitative disclosure creating ambiguity or disagreement that is reflected jointly in uncertainty and trading activity. However, because the return increment is largely uncorrelated with both dimensions, the text signal does not appear to operate primarily through common directional price expectations.

If the transcript signal simply captured good news or bad news, one would expect the return increment to move systematically with the volatility or volume increments. Instead, the strongest cross-outcome relationship is between realised variance and abnormal volume. The evidence therefore supports a view of earnings-call text as shaping how investors process and trade on information, rather than as generating a simple directional return signal. The much larger RV–VOL correlation in the text increment than in the earnings-only benchmark further suggests that this co-movement is specific to qualitative disclosure, not merely a general property of post-earnings market dynamics.

The strong contemporaneous co-movement between realised-variance and abnormal-volume text increments could reflect two different mechanisms. One possibility is pure simultaneity, where the same call content generates an immediate ambiguity shock that jointly raises uncertainty and trading activity, with no meaningful sequential structure. A second possibility is gradual information diffusion. In this interpretation, the transcript first generates uncertainty about firm value, which is then resolved through trading over subsequent days and weeks as investors process the disclosure at different speeds. Therefore, a simultaneity interpretation predicts broadly symmetric cross-outcome correlations, whereas a gradual-diffusion interpretation predicts that the uncertainty measure should lead subsequent trading activity.

I examine this distinction using three complementary tests. The first is a lead-lag asymmetry test. For each short horizon $h_s \in \{1,3,5\}$ and longer horizon $h_l \in \{1,3,5,10,15,21\}$, I compare the correlation between short-horizon text-driven realised variance and longer-horizon text-driven abnormal volume with the reverse correlation:

$$r(h_s, h_l) = Corr(\Delta_{RV}(h_s), \Delta_{VOL}(h_l)),$$

and

$$r^*(h_s, h_l) = Corr(\Delta_{VOL}(h_s), \Delta_{RV}(h_l)).$$

Under pure simultaneity, the two correlations should be similar. Under gradual diffusion, $r(\cdot)$ should exceed $r^*(\cdot)$ when $h_l > h_s$, because the early uncertainty signal should contain information about subsequent trading activity.

The second test examines whether the RV–VOL relationship is a mechanical consequence of both increments being generated from the same transcript representation For each horizon pair, I regress the realised-variance text increment and the abnormal-volume text increment separately on the corresponding text-only predictions using cross-fitted LightGBM models, and then compute the correlation between the residuals. Cross-fitting ensures that residuals are evaluated out of sample in the residualisation step. If the RV–VOL relationship is driven only by shared nonlinear exposure to the transcript embedding, the residual correlation should collapse toward zero. If it reflects a genuine uncertainty-to-trading channel, a sizeable residual correlation should remain.

The third test is a cross-sectional Granger-style exercise. For each event, I regress the longer-horizon abnormal-volume text increment on its own short-horizon value and the short-horizon realised-variance text increment:

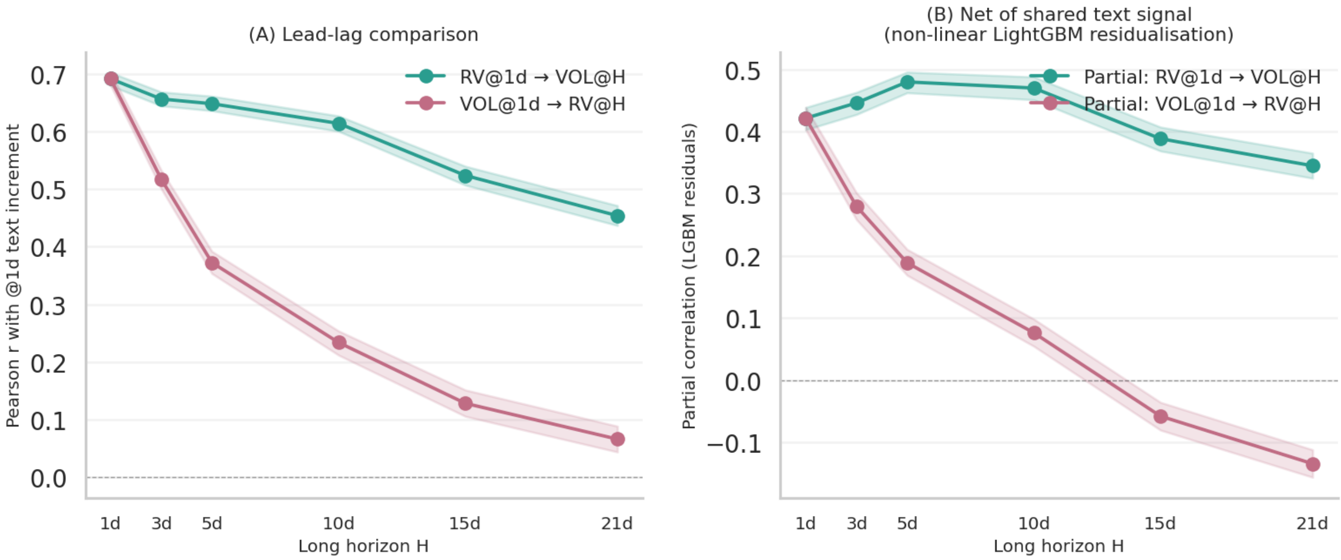
$$\Delta_{VOL}(h_l) = \alpha + \beta_1 \Delta_{VOL}(h_s) + \beta_2 \Delta_{RV}(h_s) + \varepsilon.$$

The test statistic is the incremental R^2 from adding $\Delta_{RV}(h_s)$ to a baseline specification that includes only $\Delta_{VOL}(h_s)$. This is not a time-series Granger-causality test in the strict sense, but it is a cross-sectional analogue: it asks whether text-driven uncertainty shortly after the call predicts later text-driven trading activity beyond what is already implied by early trading activity itself.

The results support the gradual-diffusion interpretation. First, the lead-lag correlations are strongly asymmetric. At the 21-day horizon, the correlation between one-day text-driven realised variance and subsequent abnormal volume is $r(1d, 21d) = 0.45$, compared with only $r^*(1d, 21d) = 0.07$ in the reverse direction. The gap widens across horizons, indicating that early text-driven uncertainty contains information about later trading activity, while early text-driven trading activity contains little information about later volatility. This asymmetry is difficult to reconcile with a pure simultaneity explanation, under which the two directions should be similar.

The nonlinear residualisation results strengthen the interpretation. For the RV-to-VOL direction, 78–81% of the raw correlation is retained at longer horizons, and the residual correlations remain highly significant across horizons. By contrast, the reverse VOL-to-RV relationship weakens substantially after nonlinear residualisation, with the retained fraction falling sharply at longer horizons. This asymmetry indicates that the sequential RV-to-VOL channel is not a mechanical artefact of shared transcript exposure. Instead, text-driven uncertainty contains information about subsequent text-driven trading activity that is not explained by the common embedding signal.

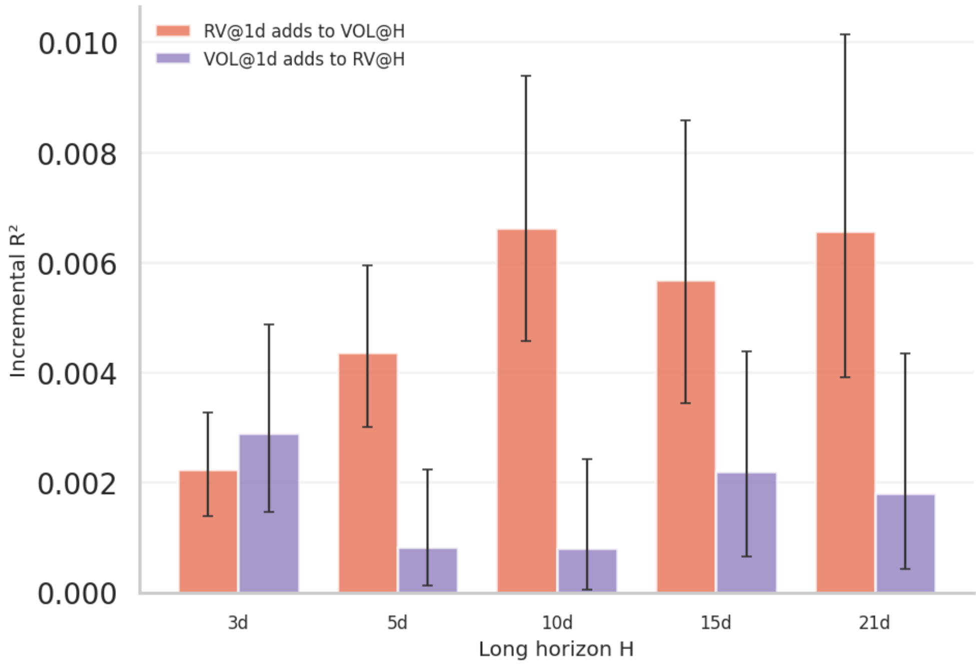
Figure 8: Lead-lag asymmetry between text-driven uncertainty and trading activity



Note: Panel (A) shows the lead-lag asymmetry between $RV@1d \rightarrow VOL@H$ and $VOL@1d \rightarrow RV@H$, with 95% bootstrap confidence intervals. Panel (B) shows raw and nonlinear-residualised correlations between $RV@1d$ and $VOL@H$, where nonlinear residuals are obtained by cross-fitted LightGBM residualisation on the corresponding text-only predictions. Confidence intervals are calculated using a permutation bootstrap.

Third, the cross-sectional Granger-style results show that short-horizon text-driven realised variance adds incremental predictive power for later abnormal volume. The coefficient on $\Delta_{RV}(1d)$ is positive and statistically significant at all longer horizons, and the incremental R^2 rises from 0.0022 at the three-day horizon to 0.0066 at the 21-day horizon. This gain is obtained after controlling for $\Delta_{VOL}(1d)$, which is itself a strong predictor of future abnormal volume. The increasing incremental contribution of early uncertainty is inconsistent with a noise-rebalancing explanation and is more consistent with qualitative disclosure generating uncertainty that is resolved gradually through trading.

Figure 9: Incremental predictive content of text-driven uncertainty for future trading



Note: The figure reports the incremental $R^2$from adding $\Delta_{RV}(1d)$ to a model using $\Delta_{VOL}(1d)$ to predict $\Delta_{VOL}(H)$. This is the cross-sectional Granger-style test. Confidence intervals are calculated using a permutation bootstrap.

The evidence suggests that earnings-call text does not merely generate a contemporaneous volume-volatility response. Rather, the text-driven realised-variance increment appears to capture an initial uncertainty shock that precedes and predicts subsequent abnormal trading activity. This strengthens the interpretation of earnings-call transcripts as a channel through which qualitative disclosure shapes the process of information absorption, as prices may incorporate directional information quickly, but uncertainty and disagreement are resolved more gradually through trading.

6. Conclusion

This paper studies the out-of-sample decay of earnings-call text informativeness across three market outcomes: abnormal returns, realised variance, and abnormal trading volume. Using a chronological train-validation-test design and sentence-embedding representations of earnings-call transcripts, the analysis examines whether qualitative disclosure improves forecasts beyond hard earnings information and standard outcome-specific controls over horizons ranging from one to sixty trading days.

The results show a consistent pattern. Hard earnings information provides essentially no incremental predictive power for abnormal returns once standard controls are included, consistent with relatively rapid price adjustment to quantitative earnings news. Earnings-call transcripts, however, contain substantial predictive content for uncertainty and trading activity.

For realised variance, text generates large short-horizon improvements, with RMSE reductions of up to roughly 8–10% at one- to three-day horizons. These gains dissipate after approximately two weeks, consistent with a short-lived uncertainty shock. For abnormal volume, the gains are larger and more persistent: RMSE reductions of around 15–18% at the one-day horizon remain positive at longer horizons, suggesting that qualitative disclosure continues to shape trading behaviour beyond the immediate call window. For abnormal returns, by contrast, the text signal is statistically detectable but economically modest, with RMSE improvements of only around 0.1–0.3% concentrated at medium horizons.

The cross-outcome results sharpen this interpretation. The text-driven increments to realised variance and abnormal volume are strongly positively correlated at the event level, with a correlation of 0.69 at the one-day horizon. By contrast, the text-driven return increment is largely orthogonal to both. This joint structure is consistent with disagreement-based models in which heterogeneous interpretation of qualitative public signals generates uncertainty and trading activity without producing a common directional price response. Further evidence shows that the realised-variance increment is the leading channel. Text-driven realised variance predicts subsequent abnormal volume, the RV-to-VOL relationship remains sizeable after controlling for the shared embedding signal, and the realised-variance increment adds Granger-style predictive content for future trading beyond early abnormal volume itself. These results suggest that earnings-call text creates ambiguity that investors resolve gradually through trading, rather than simply triggering an immediate and symmetric market response.

Several directions for future research follow naturally. First, this paper uses document-level transcript representations. Future work could exploit the internal structure of calls to identify which components generate uncertainty, such as forward guidance, risk discussion, hedging language, analyst questioning, or the distinction between prepared remarks and Q&A. Second, the analysis abstracts from firm-level heterogeneity. Characteristics such as analyst coverage, institutional ownership, retail participation, disclosure quality, and the broader information environment may influence how quickly qualitative information is processed. Estimating heterogeneous decay profiles across these dimensions would help clarify when transcript information persists longest. Third, while the results document robust predictive relationships consistent with a disagreement mechanism, they do not identify the causal effects of specific textual features. Quasi-experimental variation in call content, disclosure timing, or analyst participation could strengthen the causal interpretation.

The broader implication is that earnings-call transcripts are not primarily valuable because they generate exploitable return predictability. Their larger role is in shaping the uncertainty, attention, and trading dynamics that follow disclosure events. Qualitative text appears to generate heterogeneous interpretation, short-lived volatility, and persistent trading as ambiguity is gradually resolved. This provides a more complete account of market information absorption: prices may adjust quickly to directional news, while uncertainty and disagreement can persist beyond the initial disclosure window.

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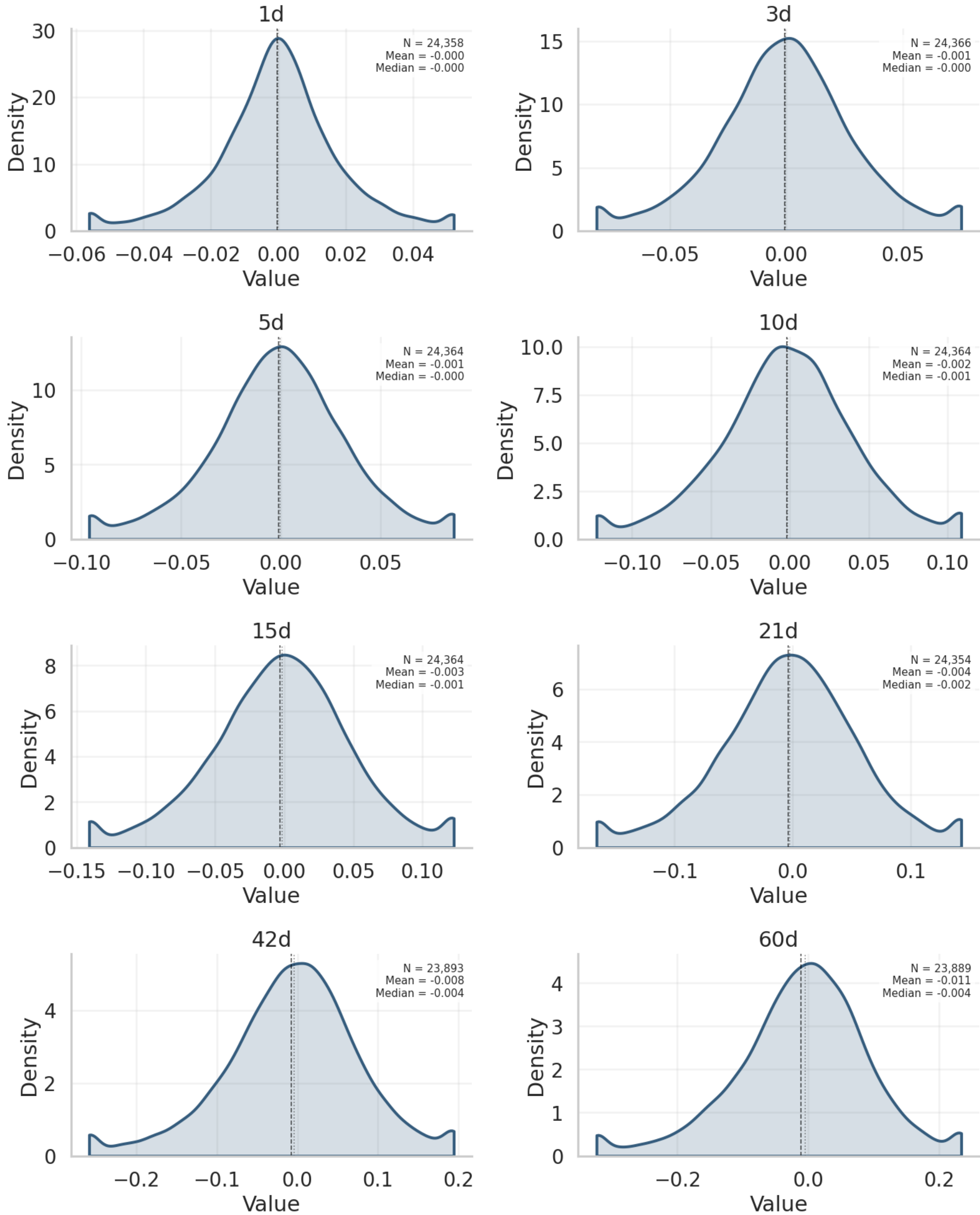
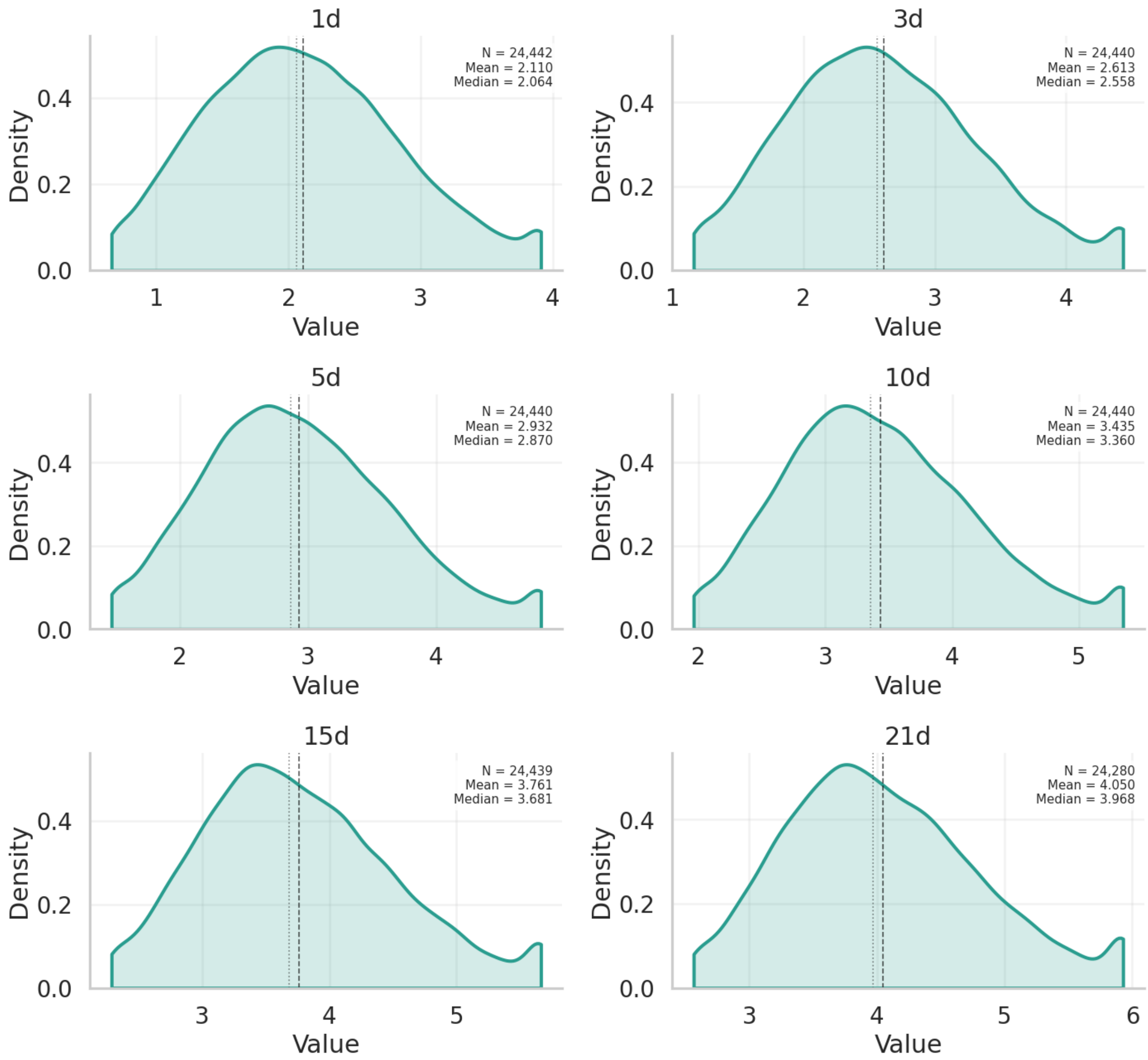
Appendix A: Additional Graphs and Tables

Figure A1: Abnormal Returns Distribution by Horizon



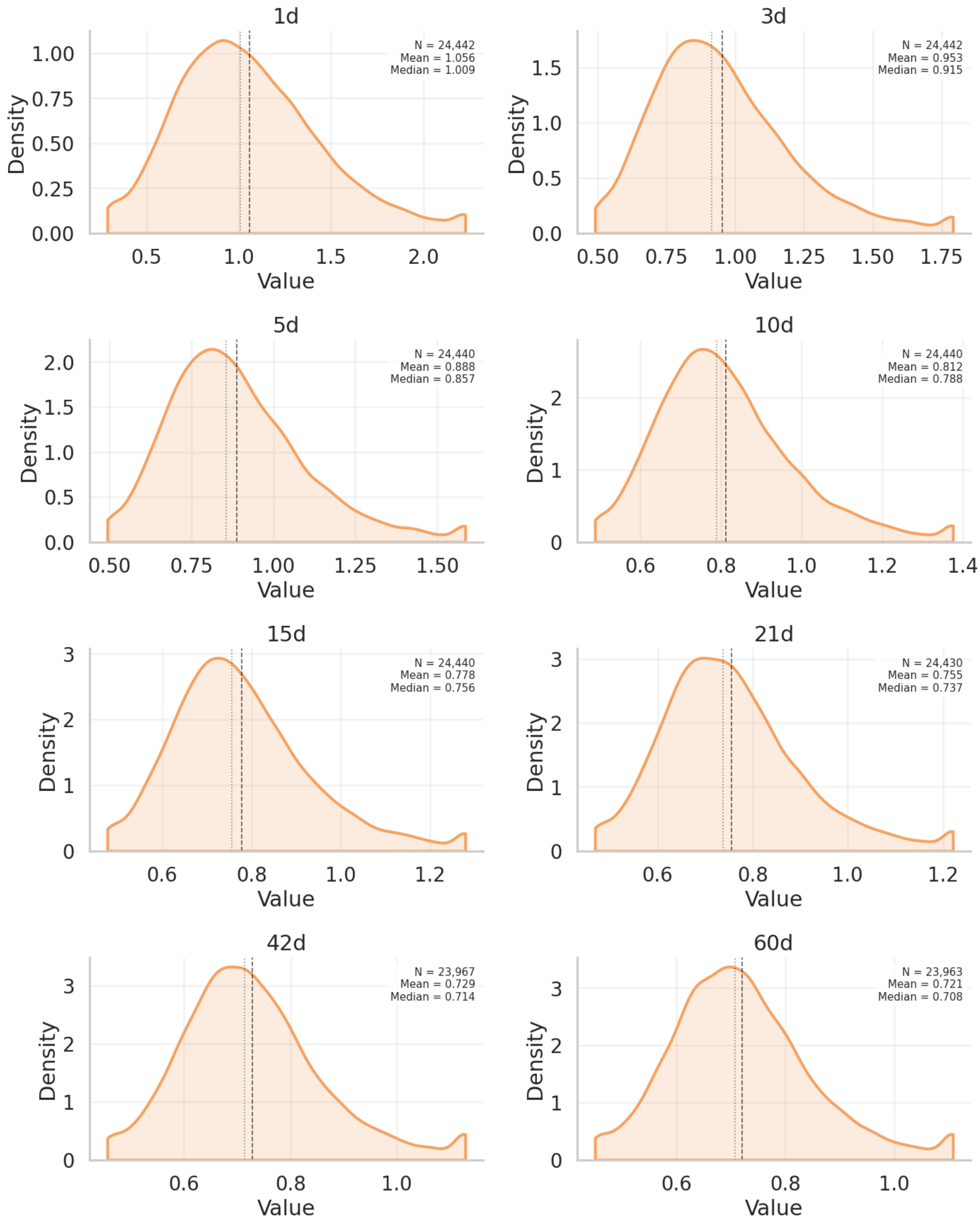
Note: Outcome winsorised at the 1st and 99th percentiles of the training sample.

Figure A2: Realised Variance Distribution by Horizon



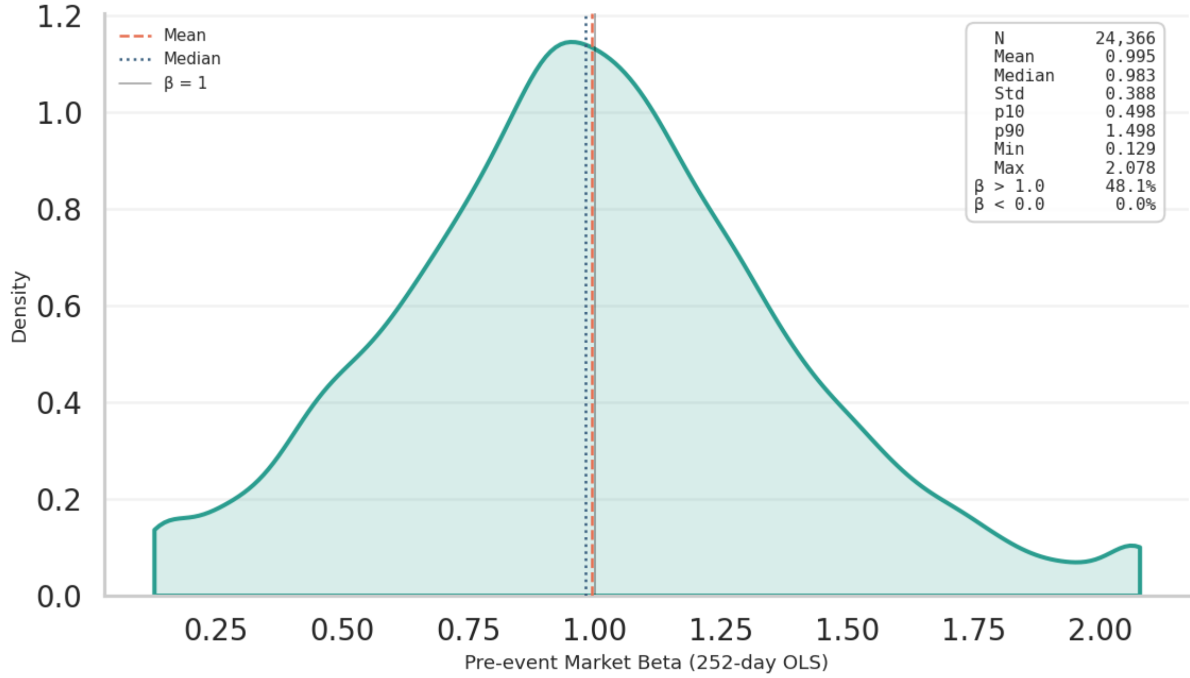
Note: Outcome winsorised at the 1st and 99th percentiles of the training sample.

Figure A3: Abnormal Volume Distribution by Horizon



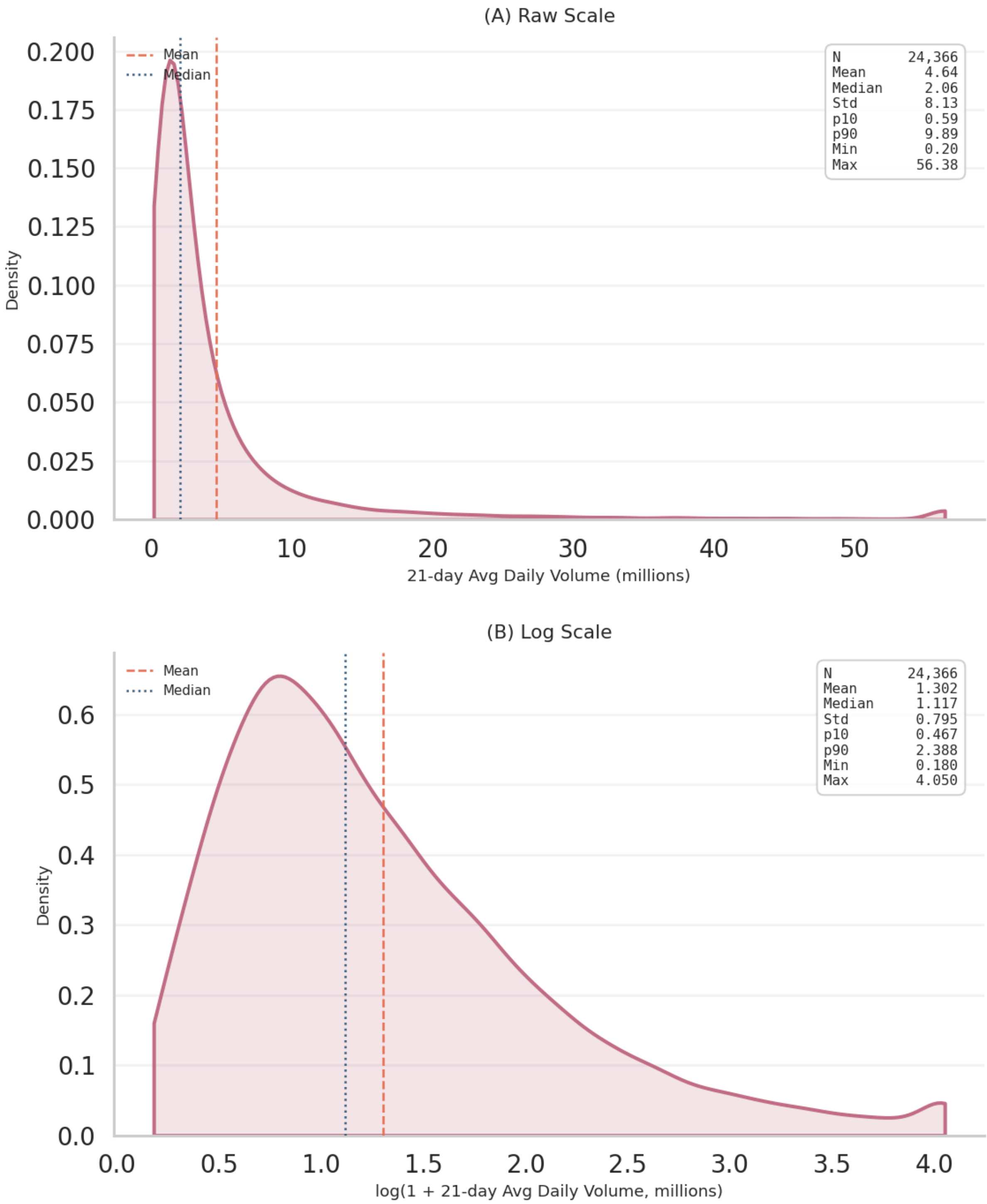
Note: Outcome winsorised at the 1st and 99th percentiles of the training sample.

Figure A4: Pre-event market β distribution



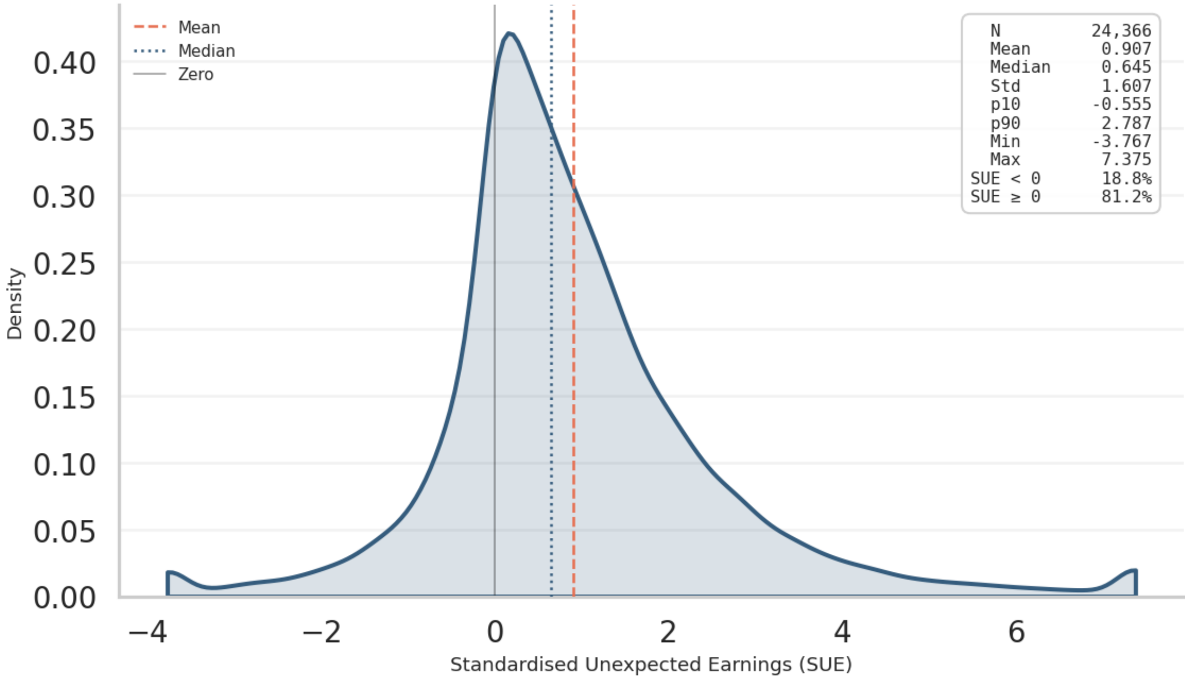
Note: The figure shows the distribution of estimated pre-event market betas for the analysis sample. Betas are estimated using firm-level daily returns and SPY returns over the pre-event estimation window. The vertical line reports the sample mean, and summary statistics are shown in the figure. The variable is winsorised at the 1st and 99th percentiles using training-sample bounds.

Figure A5: Pre-event average daily volume distribution (raw and log-scaled)



Note: The figure shows the distribution of firms' pre-event average daily trading volume, measured over the 21 trading days before the call. Panel (A) reports the raw distribution, while panel (B)reports the log-scaled distribution to account for strong right skew. Vertical lines report sample means, and summary statistics are shown in the figure. The variable is winsorised at the 1st and 99th percentiles using training-sample bounds.

Figure A6: SUE distribution



Note: The figure shows the distribution of standardised unexpected earnings (SUE), defined as reported earnings per share minus consensus analyst expectations, scaled by the firm-level historical standard deviation of earnings forecast errors. The vertical line reports the sample mean, and summary statistics are shown in the figure. The variable is winsorised at the 1st and 99th percentiles using training-sample bounds.

Table A1: Chosen parameter summary for *baseline* + *earnings* + *text* specifications

section	parameter	values
LightGBM (RV, VOL)	num_leaves	[31, 60]
	min_data_in_leaf	[20, 50]
	max_depth	[10, 20]
	feature_fraction	[0.8, 1.0]
	bagging_fraction	[0.8, 1.0]
	lambda_l1	[0.0, 0.5, 1.0]
	lambda_l2	[0.0, 1.0, 3.0]
	learning_rate	0.05 (fixed)
	early_stopping_rounds	75 (fixed)
	Total combinations	**288**
Ridge (CAR)	alpha	[0.01, 0.1, 1.0, 10.0, 100.0, 1000.0]
	Total combinations	**6**
PCA (text specs)	k (components)	[10, 20, 50, 100]

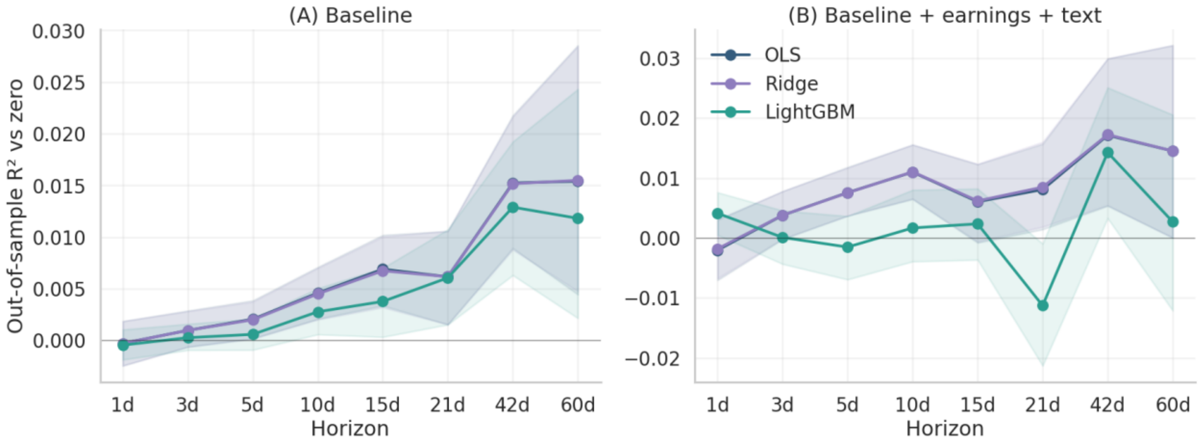
Note: PCA components are tuned jointly with model hyperparameters. The selected value of $k$is chosen using validation-sample RMSE.

Table A2: Chosen parameter summary for *baseline* + *earnings* + *text* specifications

	Model	*Horizon*	*Validation RMSE*	*PCA k*	*alpha*	*Num leaves*	*Min data in leaf*	*Max depth*	*Feature fraction*	*Bagging fraction*	*Lambda L1*	*Lambda L2*
Abnormal Returns (CAR)	RIDGE	1d	0.0203	50	1000	-	-	-	-	-	-	-
Abnormal Returns (CAR)	RIDGE	3d	0.0336	20	1000	-	-	-	-	-	-	-
Abnormal Returns (CAR)	RIDGE	5d	0.0401	20	1000	-	-	-	-	-	-	-
Abnormal Returns (CAR)	RIDGE	10d	0.0515	20	1000	-	-	-	-	-	-	-
Abnormal Returns (CAR)	RIDGE	15d	0.0617	50	1000	-	-	-	-	-	-	-
Abnormal Returns (CAR)	RIDGE	21d	0.0737	50	1000	-	-	-	-	-	-	-
Abnormal Returns (CAR)	RIDGE	42d	0.0982	20	1000	-	-	-	-	-	-	-
Abnormal Returns (CAR)	RIDGE	60d	0.1175	20	1000	-	-	-	-	-	-	-
Realised Variance (RV)	LGBM	1d	0.4721	100	-	60	20	10	1	0.8	1	3
Realised Variance (RV)	LGBM	3d	0.4546	50	-	60	20	20	0.8	0.8	1	0
Realised Variance (RV)	LGBM	5d	0.4688	50	-	31	20	10	0.8	0.8	1	3
Realised Variance (RV)	LGBM	10d	0.5409	20	-	60	20	10	0.8	0.8	0	3
Realised Variance (RV)	LGBM	15d	0.5894	100	-	60	20	10	0.8	0.8	0.5	3
Realised Variance (RV)	LGBM	21d	0.6538	50	-	60	20	10	1	0.8	1	0
Abnormal Volume (VOL)	LGBM	1d	0.3094	100	-	60	20	10	0.8	0.8	0.5	0
Abnormal Volume (VOL)	LGBM	3d	0.2204	100	-	60	50	20	1	0.8	1	3
Abnormal Volume (VOL)	LGBM	5d	0.1899	100	-	60	20	20	1	0.8	1	3
Abnormal Volume (VOL)	LGBM	10d	0.1671	100	-	31	50	20	0.8	0.8	1	3
Abnormal Volume (VOL)	LGBM	15d	0.1567	100	-	60	20	20	0.8	0.8	1	0
Abnormal Volume (VOL)	LGBM	21d	0.1531	100	-	31	20	20	1	0.8	1	0
Abnormal Volume (VOL)	LGBM	42d	0.1544	100	-	60	20	20	0.8	0.8	1	1
Abnormal Volume (VOL)	LGBM	60d	0.1496	20	-	60	20	10	0.8	0.8	1	1

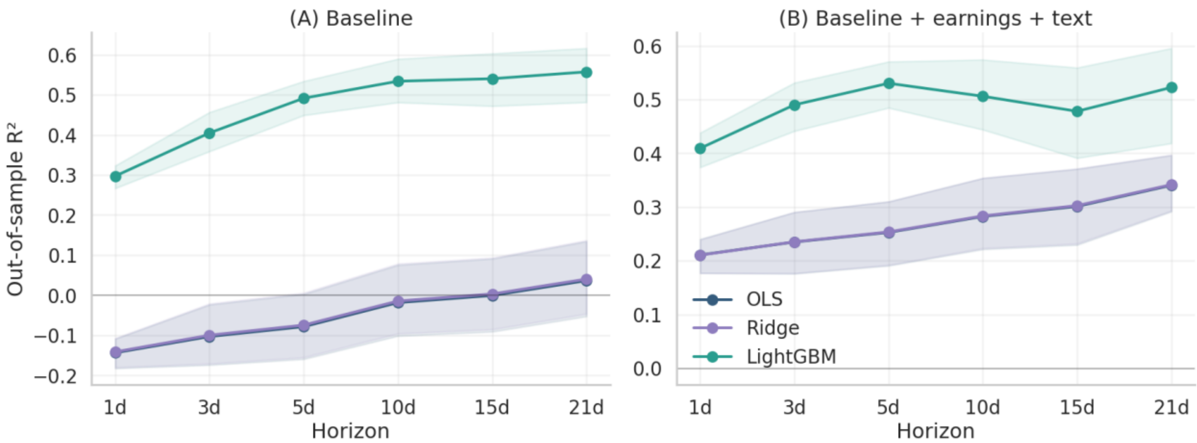
Note: PCA k denotes the number of principal components retained from the transcript embedding representation. Ridge regularisation is used for abnormal-return models, while LightGBM hyperparameters are reported for realised-variance and abnormal-volume models. Dashes indicate parameters that are not applicable to a given model family. Validation RMSE is computed on the validation sample before refitting the selected specification on the combined training and validation samples for final test-set evaluation.

Figure A7: Abnormal returns model comparison



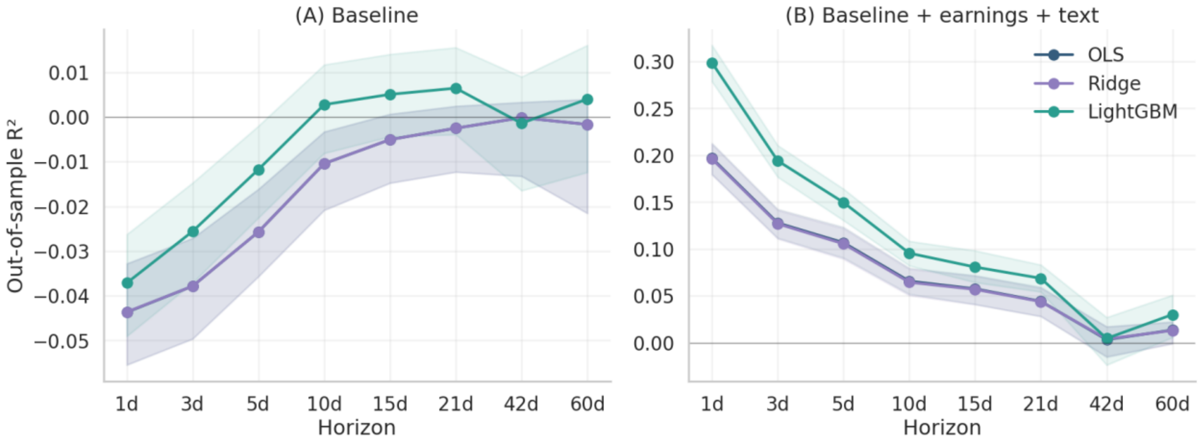
Note: The figure compares model performance across OLS, ridge, and LightGBM specifications. Confidence intervals are Holm-Bonferroni-adjusted 95% intervals calculated using the stationary bootstrap.

Figure A8: Realised variance model comparison



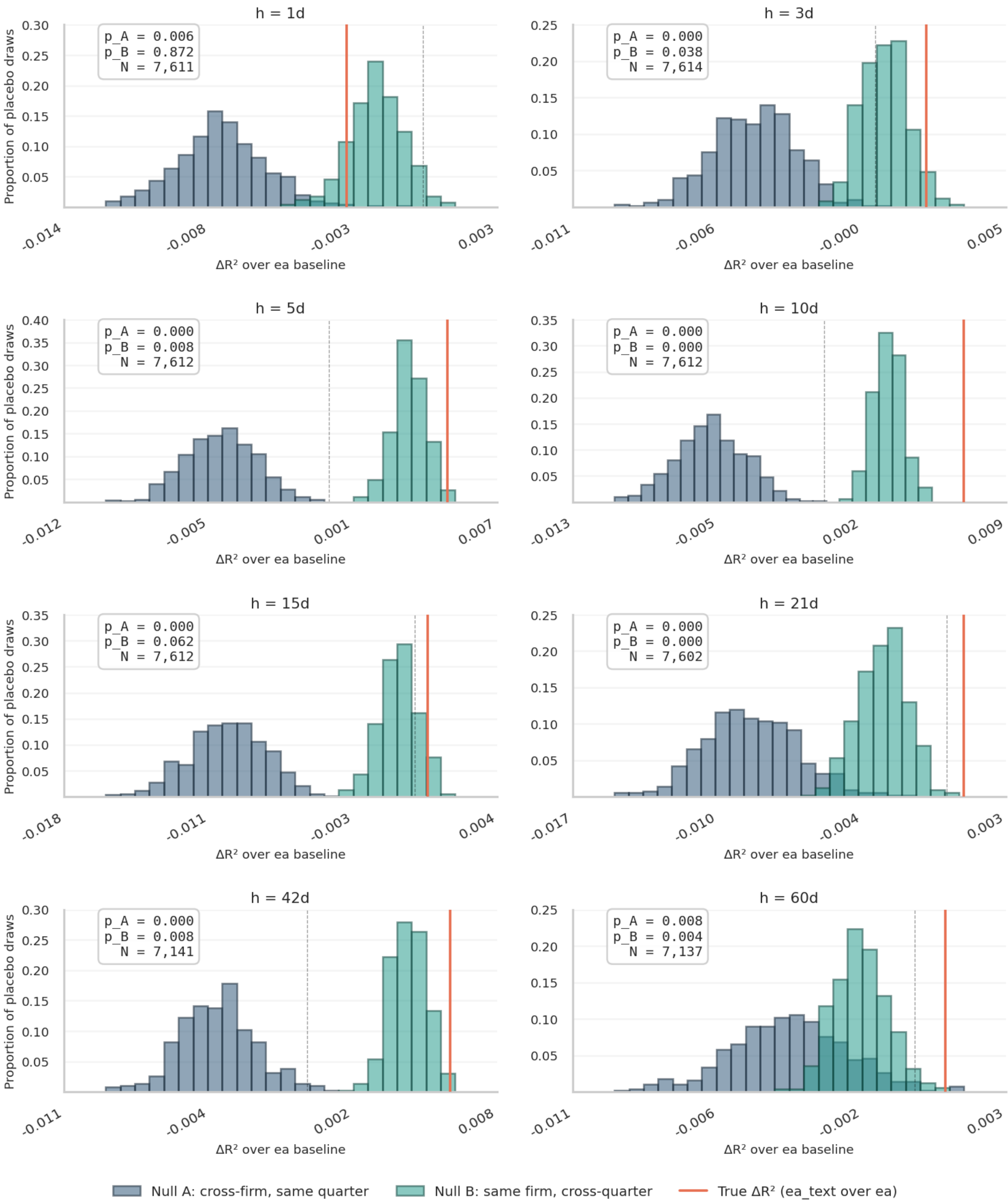
Note: The figure compares model performance across OLS, ridge, and LightGBM specifications. Confidence intervals are Holm-Bonferroni-adjusted 95% intervals calculated using the stationary bootstrap.

Figure A9: Abnormal volume model comparison



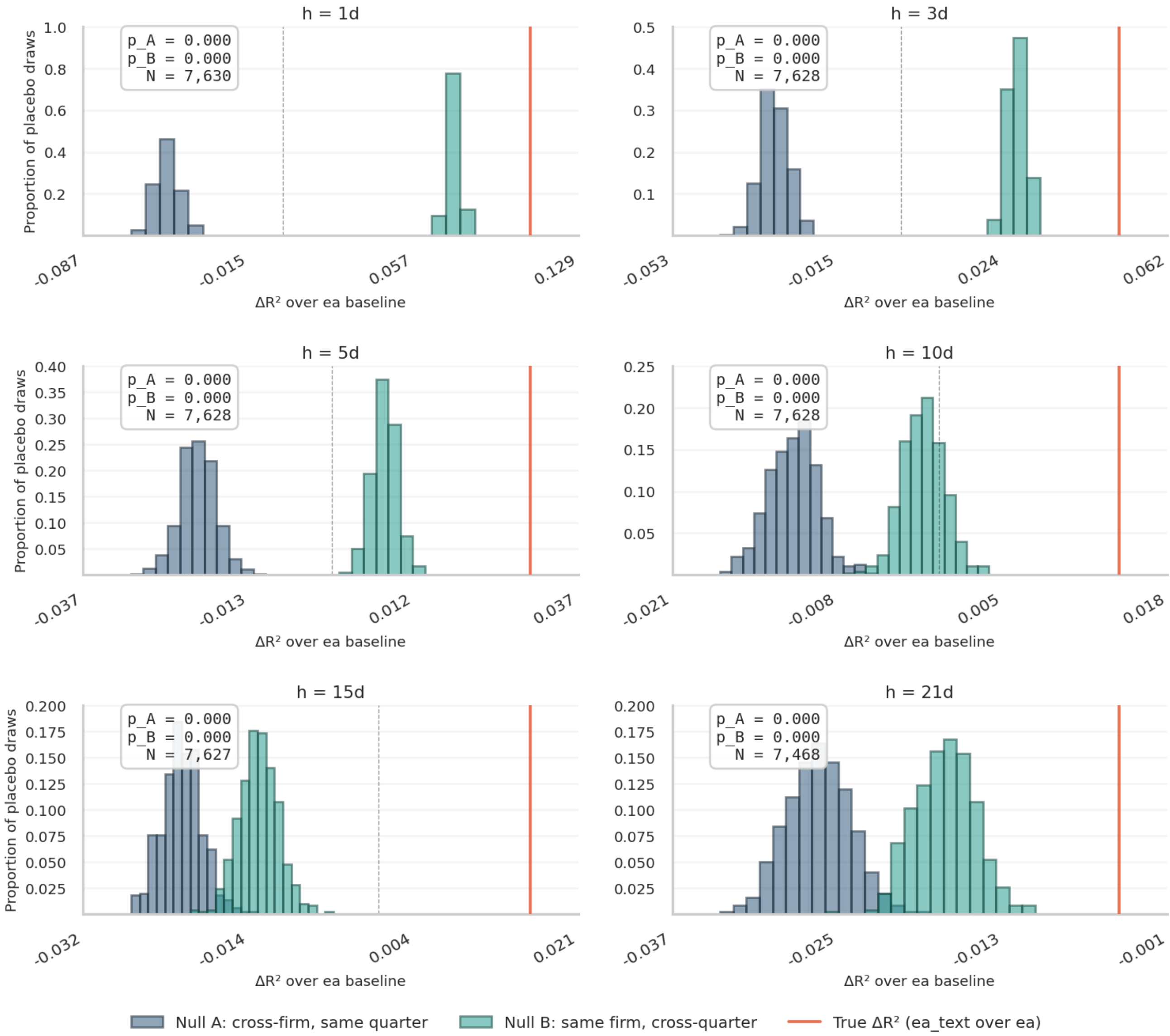
Note: The figure compares model performance across OLS, ridge, and LightGBM specifications. Confidence intervals are Holm-Bonferroni-adjusted 95% intervals calculated using the stationary bootstrap.

Figure A10: Placebo permutation test - abnormal returns



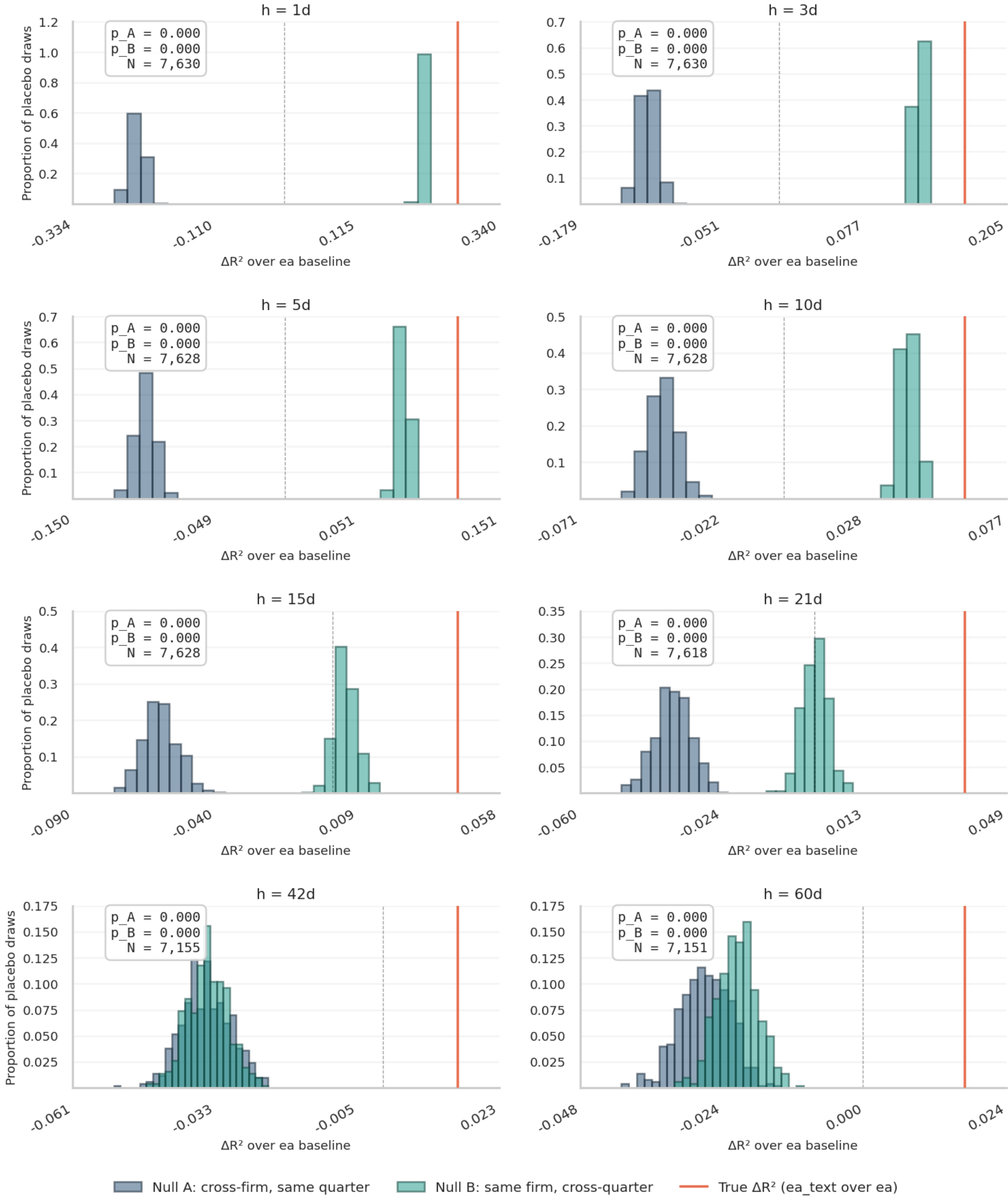
Note: Each subplot shows the null distribution of ΔR^2 under two placebo designs: cross-firm same-quarter permutation and same-firm cross-quarter permutation. The vertical line reports the true ΔR^2 from the correctly matched transcript-event pair. Distributions are probability-density estimates from 500 permutations. P-values are computed as the fraction of placebo draws with ΔR^2 greater than or equal to the true ΔR^2.

Figure A11: Placebo permutation test - realised variance



Note: Each subplot shows the null distribution of ΔR^2 under two placebo designs: cross-firm same-quarter permutation and same-firm cross-quarter permutation. The vertical line reports the true ΔR^2 from the correctly matched transcript-event pair. Distributions are probability-density estimates from 500 permutations. P-values are computed as the fraction of placebo draws with ΔR^2 greater than or equal to the true ΔR^2.

Figure A12: Placebo permutation test - abnormal volume



Note: Each subplot shows the null distribution of ΔR^2 under two placebo designs: cross-firm same-quarter permutation and same-firm cross-quarter permutation. The vertical line reports the true ΔR^2 from the correctly matched transcript-event pair. Distributions are probability-density estimates from 500 permutations. P-values are computed as the fraction of placebo draws with ΔR^2 greater than or equal to the true ΔR^2.

Table A3: Abnormal Returns Call Anchor RMSE Comparison

Horizon	*Specification A*	*Specification B*	*RMSE A*	*RMSE B*	*% Improvement*	*N*
1d	Baseline	+ Earnings	0.0209	0.0209	0.02%	7611
3d	Baseline	+ Earnings	0.0329	0.0329	0.01%	7614
5d	Baseline	+ Earnings	0.0386	0.0386	-0.02%	7612
10d	Baseline	+ Earnings	0.0479	0.0479	-0.01%	7612
15d	Baseline	+ Earnings	0.0554	0.0555	-0.03%	7612
21d	Baseline	+ Earnings	0.0639	0.0638	0.04%	7602
42d	Baseline	+ Earnings	0.0923	0.0924	-0.02%	7141
60d	Baseline	+ Earnings	0.1106	0.1105	0.01%	7137
1d	Baseline	+ Text	0.0209	0.0209	-0.09%	7611
3d	Baseline	+ Text	0.0329	0.0329	0.11%	7614
5d	Baseline	+ Text	0.0386	0.0385	0.28%**	7612
10d	Baseline	+ Text	0.0479	0.0477	0.36%***	7612
15d	Baseline	+ Text	0.0554	0.0554	0.02%	7612
21d	Baseline	+ Text	0.0639	0.0638	0.03%	7602
42d	Baseline	+ Text	0.0923	0.0921	0.29%	7141
60d	Baseline	+ Text	0.1106	0.1105	0.02%	7137
1d	Baseline	+ Earnings + Text	0.0209	0.0209	-0.13%	7611
3d	Baseline	+ Earnings + Text	0.0329	0.0329	0.10%	7614
5d	Baseline	+ Earnings + Text	0.0386	0.0385	0.25%*	7612
10d	Baseline	+ Earnings + Text	0.0479	0.0477	0.35%**	7612
15d	Baseline	+ Earnings + Text	0.0554	0.0554	0.00%	7612
21d	Baseline	+ Earnings + Text	0.0639	0.0638	0.08%	7602
42d	Baseline	+ Earnings + Text	0.0923	0.0921	0.30%	7141
60d	Baseline	+ Earnings + Text	0.1106	0.1105	0.07%	7137
1d	+ Earnings	+ Earnings + Text	0.0209	0.0209	-0.15%	7611
3d	+ Earnings	+ Earnings + Text	0.0329	0.0329	0.09%	7614
5d	+ Earnings	+ Earnings + Text	0.0386	0.0385	0.26%**	7612
10d	+ Earnings	+ Earnings + Text	0.0479	0.0477	0.36%***	7612
15d	+ Earnings	+ Earnings + Text	0.0555	0.0554	0.03%	7612
21d	+ Earnings	+ Earnings + Text	0.0638	0.0638	0.04%	7602
42d	+ Earnings	+ Earnings + Text	0.0924	0.0921	0.32%**	7141
60d	+ Earnings	+ Earnings + Text	0.1105	0.1105	0.05%	7137
1d	Baseline	Text only	0.0209	0.0209	-0.09%	7611
3d	Baseline	Text only	0.0329	0.0329	0.10%	7614
5d	Baseline	Text only	0.0386	0.0385	0.20%	7612
10d	Baseline	Text only	0.0479	0.0478	0.24%	7612
15d	Baseline	Text only	0.0554	0.0555	-0.15%	7612
21d	Baseline	Text only	0.0639	0.0639	-0.10%	7602
42d	Baseline	Text only	0.0923	0.0921	0.21%	7141
60d	Baseline	Text only	0.1106	0.1109	-0.28%	7137

Note: Holm-Bonferroni-corrected Newey-West HAC Diebold-Mariano p-values are reported using stars next to the percentage improvement. Positive values indicate lower RMSE for Specification B relative to Specification A. $* p < 0.10$, $** p < 0.05$, $*** p < 0.01$.

Table A4: Realised Variance Call Anchor RMSE Comparison

Horizon	*Specification A*	*Specification B*	*RMSE A*	*RMSE B*	*% Improvement*	*N*
1d	Baseline	+ Earnings	0.5613	0.5434	3.19%***	7630
3d	Baseline	+ Earnings	0.5000	0.4871	2.57%***	7628
5d	Baseline	+ Earnings	0.4582	0.4502	1.73%***	7628
10d	Baseline	+ Earnings	0.4318	0.4289	0.66%***	7628
15d	Baseline	+ Earnings	0.4239	0.4238	0.02%	7627
21d	Baseline	+ Earnings	0.4157	0.4134	0.55%**	7468
1d	Baseline	+ Text	0.5613	0.5095	9.24%***	7630
3d	Baseline	+ Text	0.5000	0.4730	5.40%***	7628
5d	Baseline	+ Text	0.4582	0.4412	3.71%***	7628
10d	Baseline	+ Text	0.4318	0.4271	1.08%***	7628
15d	Baseline	+ Text	0.4239	0.4216	0.54%	7627
21d	Baseline	+ Text	0.4157	0.4197	-0.97%**	7468
1d	Baseline	+ Earnings + Text	0.5613	0.4956	11.71%***	7630
3d	Baseline	+ Earnings + Text	0.5000	0.4637	7.25%***	7628
5d	Baseline	+ Earnings + Text	0.4582	0.4358	4.89%***	7628
10d	Baseline	+ Earnings + Text	0.4318	0.4213	2.41%***	7628
15d	Baseline	+ Earnings + Text	0.4239	0.4151	2.08%***	7627
21d	Baseline	+ Earnings + Text	0.4157	0.4157	-0.01%	7468
1d	+ Earnings	+ Earnings + Text	0.5434	0.4956	8.80%***	7630
3d	+ Earnings	+ Earnings + Text	0.4871	0.4637	4.81%***	7628
5d	+ Earnings	+ Earnings + Text	0.4502	0.4358	3.21%***	7628
10d	+ Earnings	+ Earnings + Text	0.4289	0.4213	1.77%***	7628
15d	+ Earnings	+ Earnings + Text	0.4238	0.4151	2.05%***	7627
21d	+ Earnings	+ Earnings + Text	0.4134	0.4157	-0.56%	7468
1d	Baseline	Text only	0.5613	0.6290	-12.06%***	7630
3d	Baseline	Text only	0.5000	0.6141	-22.82%***	7628
5d	Baseline	Text only	0.4582	0.6064	-32.35%***	7628
10d	Baseline	Text only	0.4318	0.6060	-40.36%***	7628
15d	Baseline	Text only	0.4239	0.5964	-40.70%***	7627
21d	Baseline	Text only	0.4157	0.5773	-38.89%***	7468

Note: Holm-Bonferroni-corrected Newey-West HAC Diebold-Mariano p-values are reported using stars next to the percentage improvement. Positive values indicate lower RMSE for Specification B relative to Specification A. $*\ p < 0.10$, $**\ p < 0.05$, $***\ p < 0.01$.

Table A5: Abnormal Volume Call Anchor RMSE Comparison

Horizon	*Specification A*	*Specification B*	*RMSE A*	*RMSE B*	*% Improvement*	*N*
1d	Baseline	+ Earnings	0.3952	0.3828	3.12%***	7630
3d	Baseline	+ Earnings	0.2540	0.2467	2.87%***	7630
5d	Baseline	+ Earnings	0.2092	0.2045	2.27%***	7628
10d	Baseline	+ Earnings	0.1674	0.1647	1.63%***	7628
15d	Baseline	+ Earnings	0.1512	0.1490	1.47%***	7628
21d	Baseline	+ Earnings	0.1425	0.1408	1.18%***	7618
42d	Baseline	+ Earnings	0.1264	0.1252	0.90%***	7155
60d	Baseline	+ Earnings	0.1233	0.1223	0.80%***	7151
1d	Baseline	+ Text	0.3952	0.3310	16.23%***	7630
3d	Baseline	+ Text	0.2540	0.2285	10.04%***	7630
5d	Baseline	+ Text	0.2092	0.1935	7.51%***	7628
10d	Baseline	+ Text	0.1674	0.1614	3.61%***	7628
15d	Baseline	+ Text	0.1512	0.1470	2.80%***	7628
21d	Baseline	+ Text	0.1425	0.1394	2.23%***	7618
42d	Baseline	+ Text	0.1264	0.1250	1.11%***	7155
60d	Baseline	+ Text	0.1233	0.1218	1.24%***	7151
1d	Baseline	+ Earnings + Text	0.3952	0.3243	17.93%***	7630
3d	Baseline	+ Earnings + Text	0.2540	0.2242	11.74%***	7630
5d	Baseline	+ Earnings + Text	0.2092	0.1910	8.68%***	7628
10d	Baseline	+ Earnings + Text	0.1674	0.1592	4.91%***	7628
15d	Baseline	+ Earnings + Text	0.1512	0.1456	3.72%***	7628
21d	Baseline	+ Earnings + Text	0.1425	0.1380	3.19%***	7618
42d	Baseline	+ Earnings + Text	0.1264	0.1243	1.65%***	7155
60d	Baseline	+ Earnings + Text	0.1233	0.1213	1.68%***	7151
1d	+ Earnings	+ Earnings + Text	0.3828	0.3243	15.29%***	7630
3d	+ Earnings	+ Earnings + Text	0.2467	0.2242	9.13%***	7630
5d	+ Earnings	+ Earnings + Text	0.2045	0.1910	6.56%***	7628
10d	+ Earnings	+ Earnings + Text	0.1647	0.1592	3.33%***	7628
15d	+ Earnings	+ Earnings + Text	0.1490	0.1456	2.28%***	7628
21d	+ Earnings	+ Earnings + Text	0.1408	0.1380	2.03%***	7618
42d	+ Earnings	+ Earnings + Text	0.1252	0.1243	0.76%**	7155
60d	+ Earnings	+ Earnings + Text	0.1223	0.1213	0.88%***	7151
1d	Baseline	Text only	0.3952	0.3366	14.83%***	7630
3d	Baseline	Text only	0.2540	0.2313	8.95%***	7630
5d	Baseline	Text only	0.2092	0.1977	5.51%***	7628
10d	Baseline	Text only	0.1674	0.1632	2.53%***	7628
15d	Baseline	Text only	0.1512	0.1487	1.62%***	7628
21d	Baseline	Text only	0.1425	0.1414	0.81%**	7618
42d	Baseline	Text only	0.1264	0.1261	0.23%	7155
60d	Baseline	Text only	0.1233	0.1237	-0.33%	7151

Note: Holm-Bonferroni-corrected Newey-West HAC Diebold-Mariano p-values are reported using stars next to the percentage improvement. Positive values indicate lower RMSE for Specification B relative to Specification A. $* \, p < 0.10$, $** \, p < 0.05$, $*** \, p < 0.01$.

Appendix B: The economic value of earnings text

The main results show that earnings-call text generates statistically significant improvements in out-of-sample forecasting performance. This appendix asks whether these forecast improvements correspond to economically interpretable cross-sectional signals. The exercises are not intended to establish tradable profitability. They use paper returns at the event level and do not optimise transaction costs, position sizing, leverage, liquidity, short-sale constraints, or overlapping holding periods. Even modest execution costs could substantially reduce the reported Sharpe differentials. The purpose is therefore to assess whether the forecast improvements reflect meaningful ranking power consistent with limits to arbitrage and gradual information absorption, rather than to propose an implementable trading strategy.

I present two validation exercises. The first asks whether the text-augmented return prediction generates a stronger long-short portfolio signal than the earnings-only prediction. The second asks whether the text-augmented realised-variance prediction produces better-calibrated Value at Risk estimates around earnings calls.

B.1. Return predictions: long-short quintile strategy

For each call-anchor horizon h, test-set events are ranked into five equal-sized quintiles using the text-augmented abnormal-return prediction, $\hat{y}^{EA+Text}(h)$. A paper long-short portfolio goes long the top quintile, Q5, and short the bottom quintile, Q1, with equal weights within each leg. The same procedure is applied using the earnings-only prediction, $\hat{y}^{EA}(h)$, as the ranking signal. The annualised Sharpe ratio is computed as

$$SR_{ann} = SR_{raw} \times \sqrt{252/h},$$

where SR_{raw} is the mean divided by the standard deviation of event-level long-short returns. The primary statistic is the difference in annualised Sharpe ratios between the text-augmented and earnings-only strategies. Statistical significance is assessed using 1,000 bootstrap draws with replacement.

At the one-day horizon, the two strategies perform similarly, with annualised Sharpe ratios of 0.51 for the text-augmented ranking and 0.56 for the earnings-only ranking. This is consistent with the main forecasting result that transcript text adds little directional return content immediately after the call. The text-augmented strategy begins to outperform at medium horizons. At five days, its annualised Sharpe ratio is 0.73, compared with 0.45 for the earnings-only strategy, a difference of 0.28 with a bootstrap p-value below 0.10. At ten days, the difference is again 0.28 and is statistically significant at the 5% level. The advantage fades at 15- and 21-day horizons and reappears weakly at 42 days.

Table B1: CAR long-short strategy performance by horizon

Horizon	N	SR(ea+text)	SR(ea)	ΔSharpe	Q5(ea+text)	Q1(ea+text)	L/S(ea+text)
1d	7611	0.511	0.558	-0.047	0.0005	-0.0009	0.0014
3d	7614	0.653	0.381	0.272	0.0014	-0.0035	0.0048
5d	7612	0.731	0.447	0.284*	0.0029	-0.0051	0.008
10d	7612	0.701	0.417	0.284**	0.0041	-0.0095	0.0135
15d	7612	0.474	0.458	0.016	0.0035	-0.0099	0.0133
21d	7602	0.402	0.41	-0.008	0.0022	-0.0132	0.0154
42d	7141	0.326	0.24	0.086*	0.002	-0.0239	0.0259
60d	7137	0.237	0.201	0.036	0.0002	-0.0279	0.0281

Note: The table reports annualised Sharpe ratios for the $EA + Text$-ranked and EA-only-ranked long-short strategies, the Sharpe ratio difference, and mean realised abnormal returns for Q5 and Q1 under the text-augmented ranking. The final column reports the Q5–Q1 spread. Portfolios are equal-weighted paper portfolios with no transaction costs. Sharpe ratios are annualised as $SR_{raw} \times \sqrt{252/h}$. $* \, p < 0.10$, $** \, p < 0.05$, $*** \, p < 0.01$.

This horizon profile closely matches the main return-forecasting results, where text-driven improvements are also concentrated at five- to ten-day horizons. The exercise therefore shows that the modest RMSE gains for abnormal returns correspond to meaningful portfolio-level discrimination at the horizons where the text signal is strongest, even though the overall return-predictive content remains economically limited.

B.2. Realised variance: Value at Risk backtesting

The main results show that transcript text generates large short-horizon improvements in realised-variance forecasting. A natural economic application is whether these improvements translate into better-calibrated risk measures around earnings calls. I test this by using the text-augmented and earnings-only realised-variance predictions to construct parametric Value at Risk estimates for each event, and I compare violation rates using the Kupiec (1995) proportion-of-failures test.

For each test-set event at horizon h, the predicted h-day cumulative realised variance, $\hat{y}_{RV}(h)$, is converted into a daily volatility estimate:

$$\hat{\sigma} = \sqrt{\frac{\exp\left(\hat{y}_{RV}\right) - 1}{h \times 10000}}.$$

The division by 10,000 reverses the percentage-point scaling of the five-minute returns. The h-day VaR at confidence level α is then

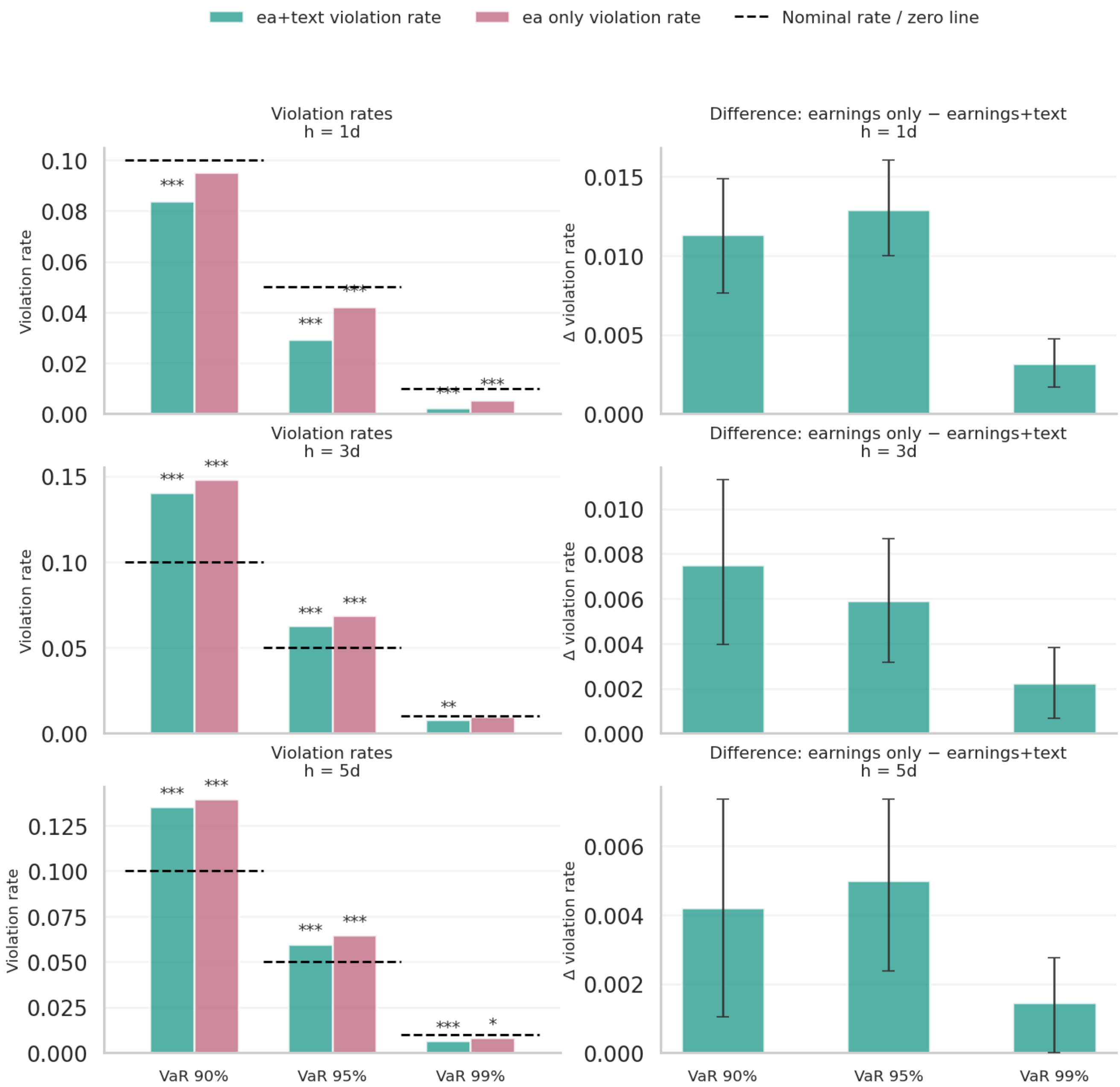
$$VaR_{\alpha,h} = z_\alpha \hat{\sigma} \sqrt{h},$$

where z_α is the relevant normal quantile. A violation occurs when the absolute realised cumulative abnormal return exceeds the VaR threshold:

$$|CAR(h)| > VaR_{\alpha,h}.$$

A well-calibrated model should produce a violation rate close to the nominal tail probability, $1 - \alpha$. I compare violation rates at 90%, 95%, and 99% confidence levels for one-day, three-day, and five-day horizons.

Figure B1: VaR backtesting results



Note: The top row shows actual violation rates at 90%, 95%, and 99% VaR. Dashed lines indicate nominal violation rates. Kupiec stars indicate rejection of correct unconditional coverage, where rejection implies miscalibration. The bottom row shows the difference in violation rates, defined as EA-only minus EA + Text, with 95% bootstrap confidence intervals. Positive values indicate fewer violations under the text-augmented model. $*$ $p < 0.10$, $**$ $p < 0.05$, $***$ $p < 0.01$.

The results show a clear pattern at the one-day horizon. For 90% VaR, the earnings-only model is reasonably well calibrated, with an actual violation rate of 9.5% against a nominal rate of 10.0%. The text-augmented model produces a lower violation rate of 8.4%. For 95% VaR, the earnings-only model produces a violation rate of 4.2%, while the text-augmented model produces 2.9%. For 99% VaR, the corresponding violation rates are 0.5% and 0.2%. Thus, the text-augmented model produces fewer violations across all confidence levels at the one-day horizon.

The interpretation requires care. Both models generate violation rates below the nominal tail probabilities at the one-day horizon, implying conservative VaR estimates under the parametric normal approximation. This may reflect the fact that realised post-call abnormal returns in the test sample are less extreme than implied by a normal VaR model calibrated from realised-variance forecasts, or that the model is better suited to ranking risk than to exact distributional calibration. The text-augmented model is even more conservative than the earnings-only model, because it predicts higher event-level volatility for observations identified as high uncertainty, leading to wider VaR thresholds and fewer violations.

This does not imply that the text-augmented model delivers tighter risk limits. Rather, it suggests that transcript text improves the identification of events with elevated short-run risk. From a risk-management perspective, this is still economically meaningful, as the text-augmented model flags high-uncertainty earnings calls more effectively and produces fewer VaR breaches around those events. The result is therefore best interpreted as evidence that text improves event-level risk discrimination, not as evidence of perfectly calibrated or more capital-efficient VaR estimates.

At the three- and five-day horizons, both models tend to produce more violations than expected, indicating weaker calibration over longer holding periods. This is consistent with the main forecasting results, where the incremental volatility-predictive value of earnings-call text is strongest immediately after the call and dissipates quickly. The text-augmented model continues to produce fewer violations than the earnings-only model at these horizons, although the gap is smaller than at one day.

Appendix C: Temporal stability and market regimes

The main results pool all test-set events from June 2021 to May 2025. This period spans materially different macro-financial conditions, including the post-pandemic monetary tightening cycle, elevated inflation, and subsequent disinflation. I therefore examine whether the text-driven forecasting gains are stable across market regimes or instead concentrated in a specific subperiod. The exercise uses two splits of the held-out test sample and does not refit any models, with all predictions coming directly from the primary out-of-sample evaluation.

The first split uses the level of the VIX index on the call date, obtained from Yahoo Finance[12]. Events are classified as high-VIX if the VIX index on the call date is at or above the test-sample median of 18.5, and low-VIX otherwise. This tests whether the forecasting gains from transcript text are concentrated in periods of elevated aggregate uncertainty. The second split divides the test period at its chronological midpoint, creating two subsamples: June 2021 to May 2023 and June 2023 to May 2025. This tests whether the text signal is stable across the early and late portions of the evaluation window.

The results show that the text-driven gains are not concentrated in a single market regime. For realised variance, the text signal is positive in both VIX regimes and both calendar halves at short horizons, with one-day ΔR^2 values ranging from 0.073 to 0.147. The gains are somewhat larger in low-VIX environments and in the second half of the test period, suggesting that the transcript signal is not merely a crisis-period artefact. At the 21-day horizon, however, ΔR^2 is slightly negative in some subsamples, consistent with the main result that volatility-related text information dissipates after approximately two weeks.

For abnormal volume, the text-driven ΔR^2 is positive in every subsample and at every horizon, ranging from 0.010 to 0.309. The volume signal is therefore not confined to high-uncertainty periods or to a specific calendar episode. If anything, the gains are larger in low-VIX and later-period subsamples, suggesting that the text-driven attention channel is a persistent feature of earnings-call information processing rather than a stress-period phenomenon. For abnormal returns, the pattern is also consistent with the main results: gains are most visible at the medium horizons, particularly five and ten trading days, but remain smaller and noisier than the realised-variance and abnormal-volume effects.

In summary, the subperiod analysis indicates that the main results are not driven by a particular market environment or a narrow episode within the test window. The same qualitative pattern appears across both regime splits, with transcript text generating large and persistent gains for abnormal volume, meaningful short-horizon gains for realised variance that decay after roughly two weeks, and modest return-predictive gains concentrated at medium horizons.

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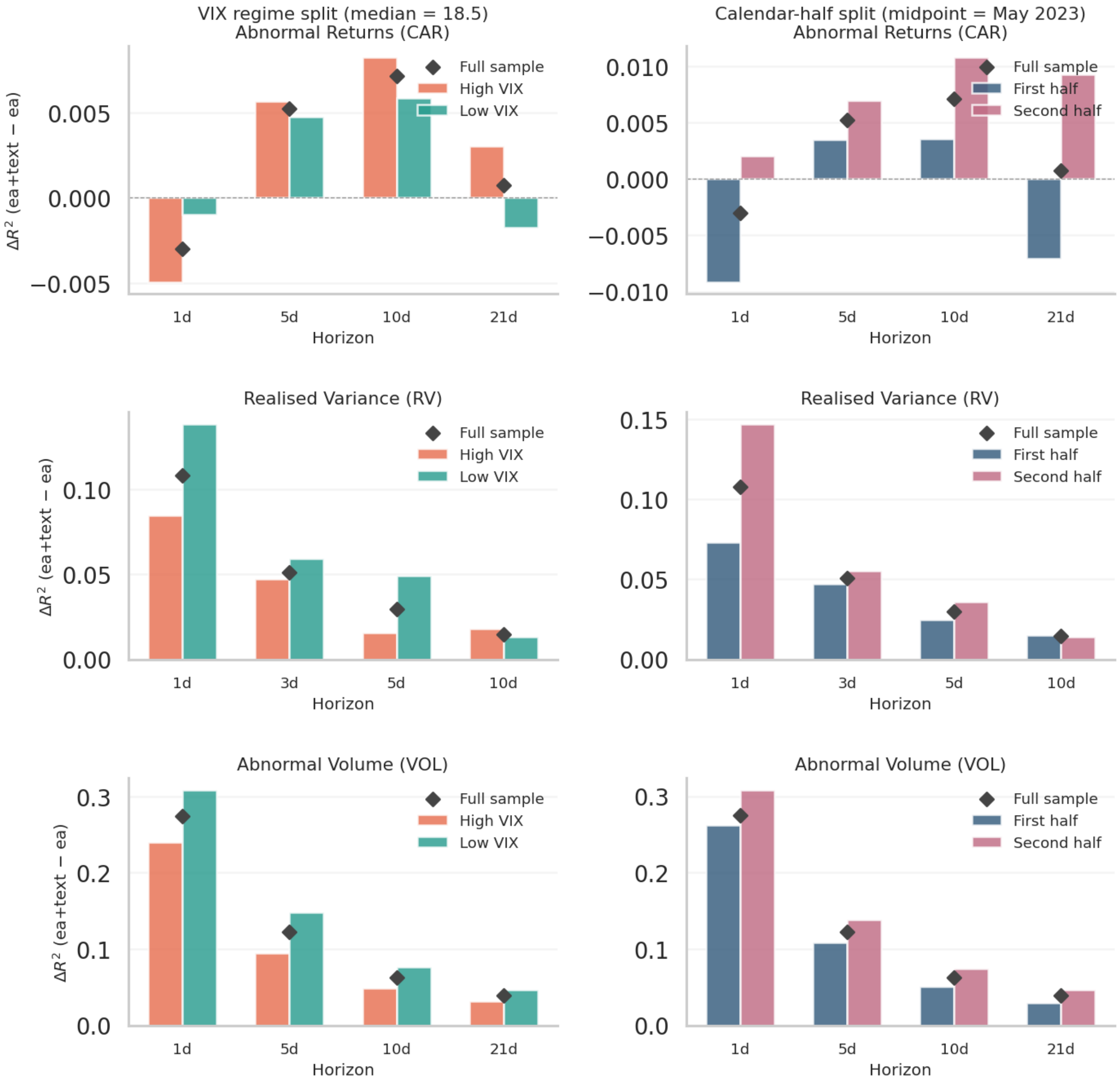
[12] Obtained through the `yfinance` API: https://ranaroussi.github.io/yfinance/

Figure C1: Temporal stability of text-driven forecasting gains



Note: The figure reports ΔR^2 for the text-augmented model relative to the earnings-only model across test-sample subsamples. The top row splits events by the VIX index on the call date, using the test-sample median of 18.5 as the cutoff. The bottom row splits the test period into two calendar halves: June 2021–May 2023 and June 2023–May 2025. Diamond markers show the full test-sample ΔR^2. Models are not refit within subsamples; all predictions are from the primary chronological out-of-sample evaluation.

Appendix D: Embedding representation

Large language models represent text numerically, allowing semantic relationships between documents to be analysed using vector operations. In an embedding model, a sentence or document is mapped into a dense vector in a high-dimensional space. The model is trained so that texts with similar meanings tend to lie closer together, while semantically different texts are placed further apart. As a result, two statements can be close in embedding space even if they share little vocabulary.

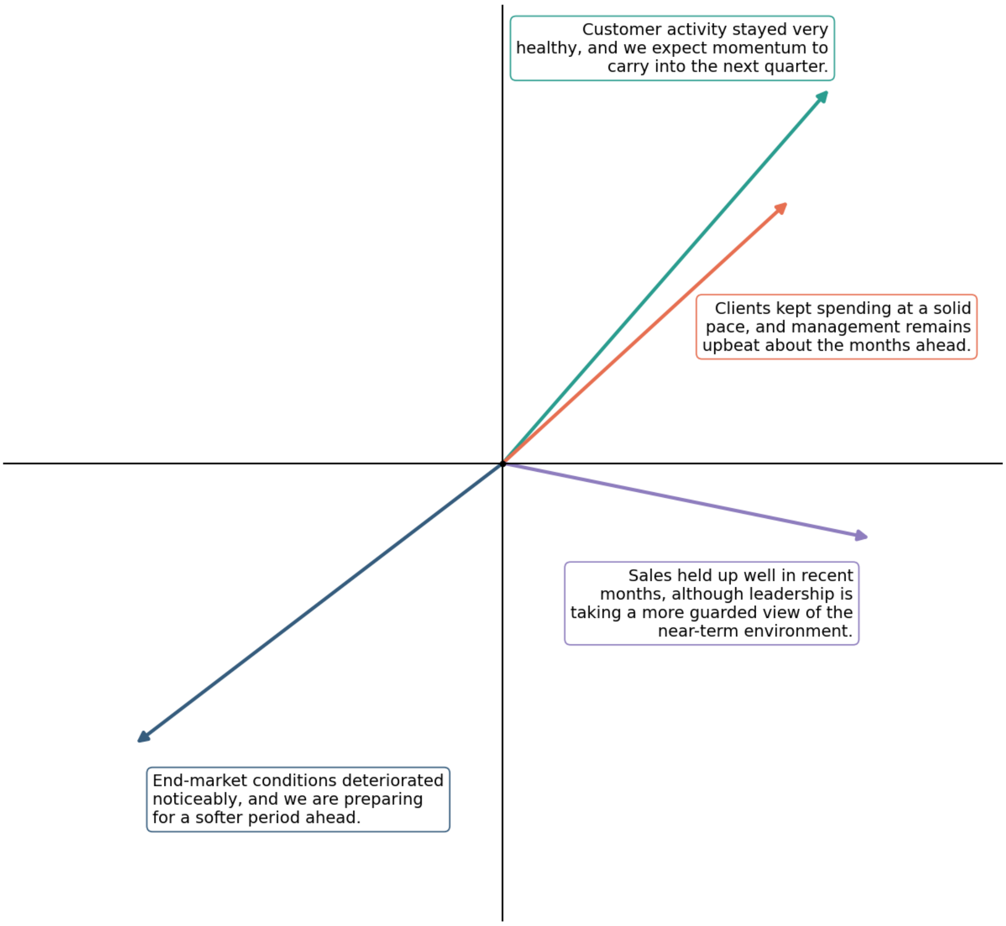
For example, the statements “Customer activity stayed very healthy, and we expect momentum to carry into the next quarter” and “Clients kept spending at a solid pace, and management remains upbeat about the months ahead” would be expected to have similar embeddings because they convey similar information about current trading conditions and the near-term outlook. This is one advantage of embedding-based representations over bag-of-words or dictionary-based methods, which rely more heavily on word overlap and fixed vocabulary lists.

Transformer-based embedding models construct these representations by contextualising each token in a sequence, so that the representation of a word depends on the surrounding words. A sentence embedding is then formed by pooling these contextualised token representations into a fixed-length vector. In practice, the individual coordinates of the embedding are not directly interpretable. Meaning is distributed across many dimensions, and semantic relationships are captured through the geometry of the embedding space (for example, through distances, angles, and relative positions).

Appendix Figure D1 provides a stylised two-dimensional illustration of this logic. The example is purely illustrative, as actual embeddings in this paper are 768-dimensional and the axes do not correspond to labelled economic concepts. The purpose of the figure is simply to show that texts can be represented as points in a learned semantic space, where proximity reflects similarity in meaning rather than exact word overlap.

In the empirical analysis, embeddings are generated using **`BAAI/bge-base-en-v1.5`**. Each transcript is split into sentences, each sentence is embedded into a 768-dimensional vector, and the transcript-level representation is constructed by averaging across sentence embeddings. The resulting vector is standardised and reduced using principal components before entering the forecasting models. Principal components are always estimated only on the training sample and then applied to the validation and test samples to avoid look-ahead bias.

Figure D1: Illustration of embedding-space geometry in low dimensions



Note: The figure provides a stylised low-dimensional illustration of embedding geometry. Actual transcript embeddings used in the analysis are 768-dimensional, and individual coordinates are not directly interpretable.